

Computer Graphics

COMP3421 and COMP9415

Never Stand Still

Introduction to Unreal Engine

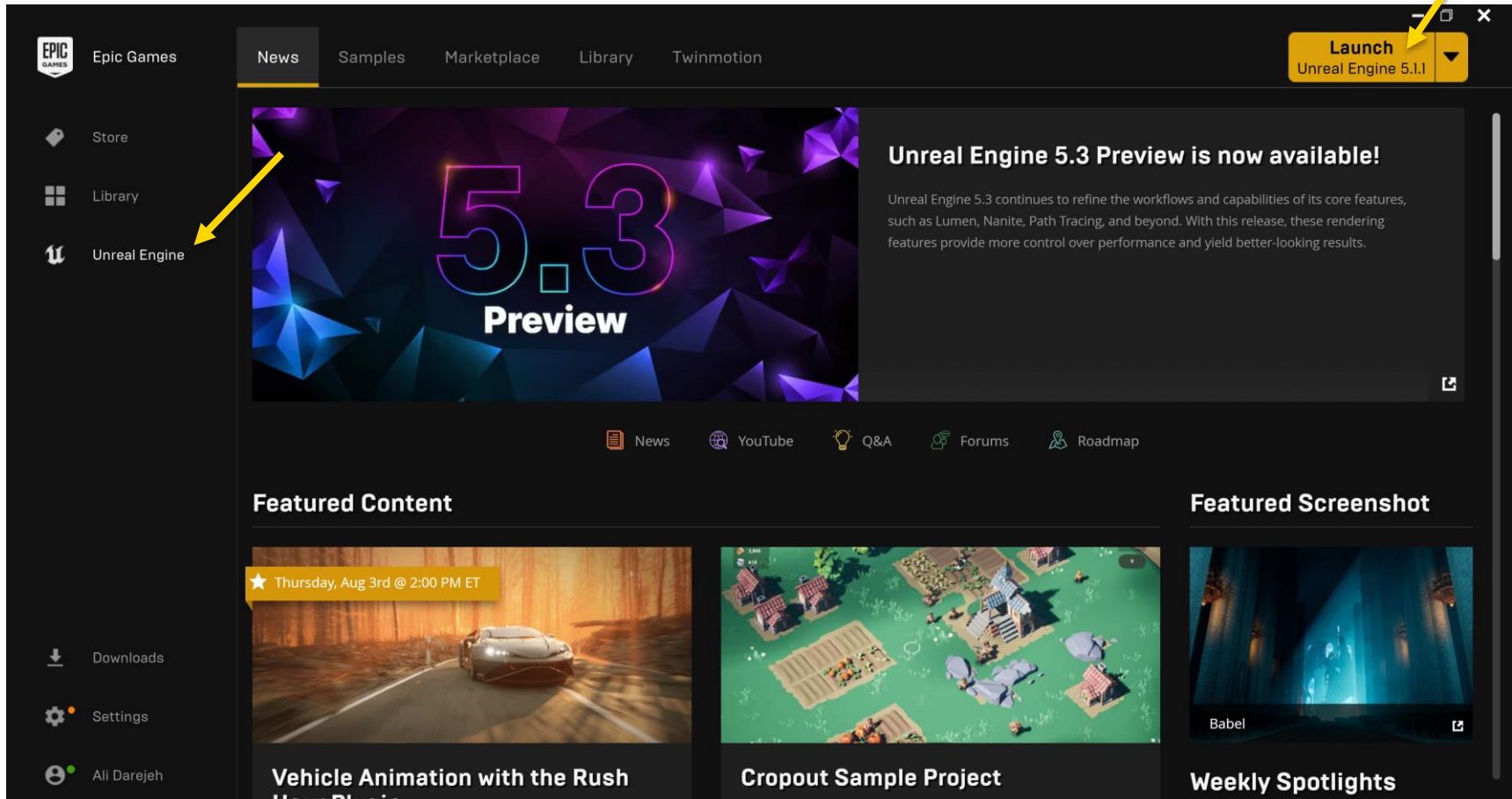
Dr Ali Darejeh
School of Computer Science and Engineering
Term 3, 2023

Content overview

- Introduction to Unreal Engine 5
- C++ versus Blueprint visual scripting
- Unreal Engine asset store
- Transforming objects in 3D
- Introduction to blueprints

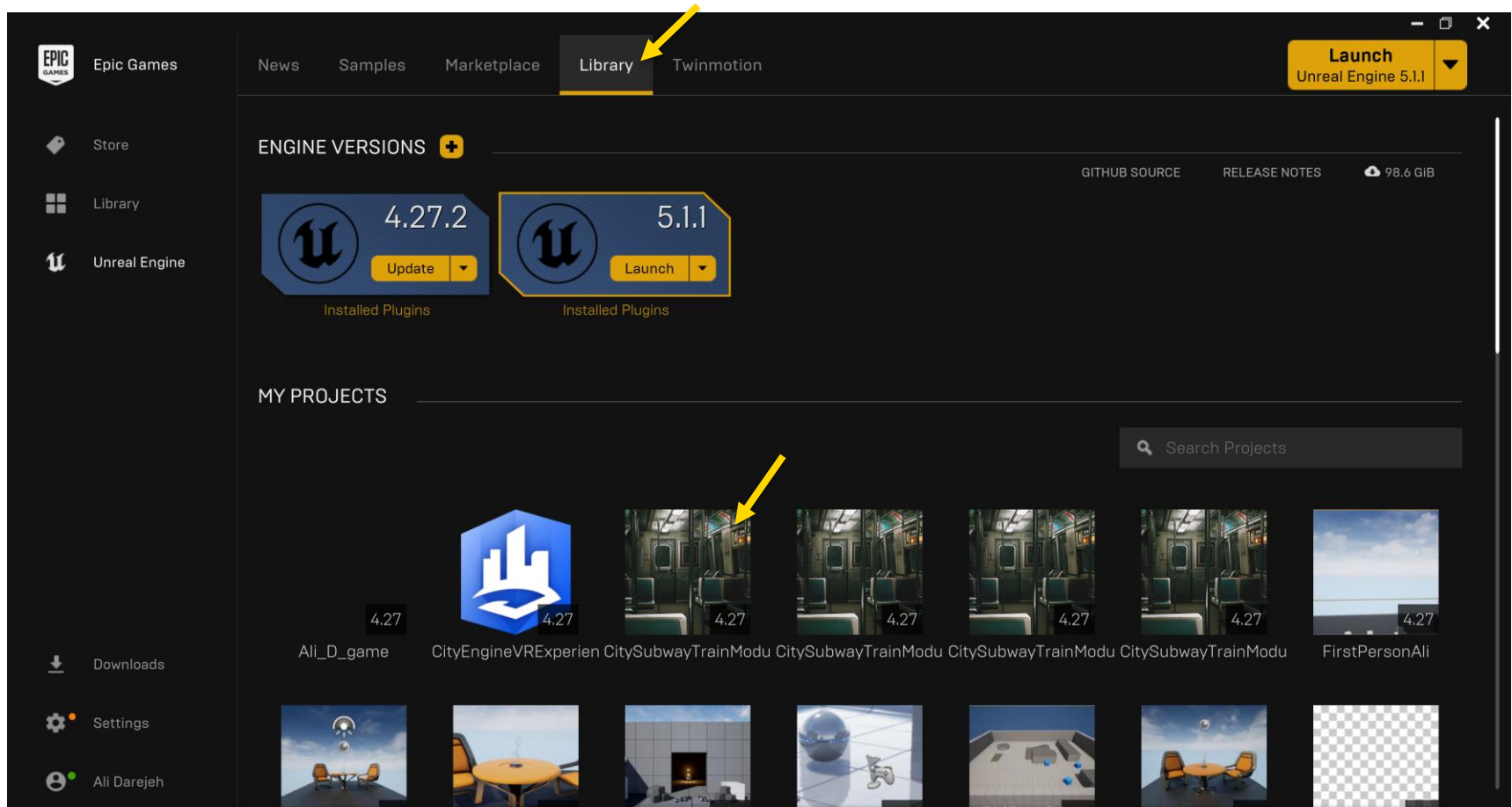
Create a new project

- Open Unreal Engine launcher > Click Unreal Engine tab > Click the launcher button > Select the desired engine version.



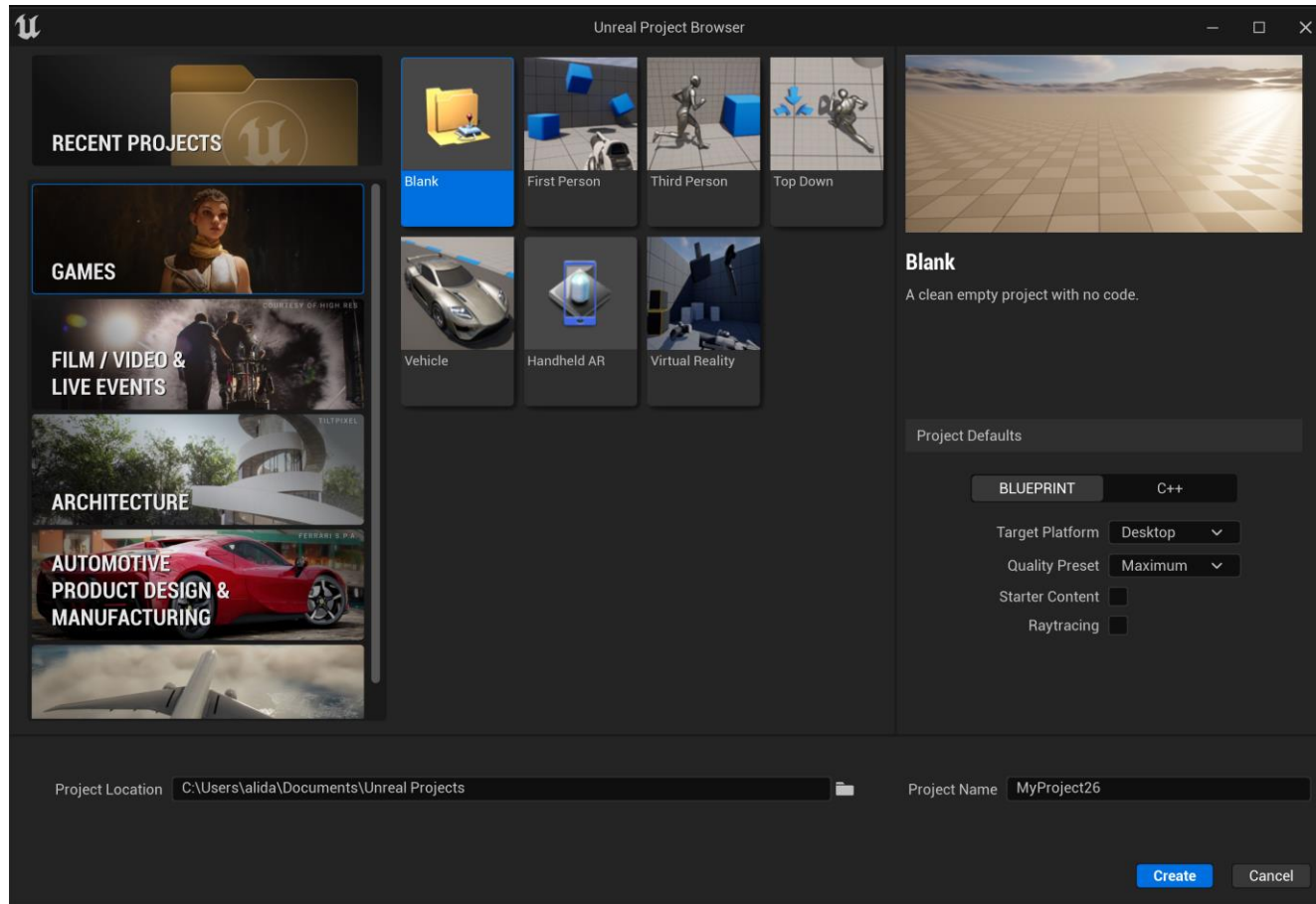
Open an existing project

- Open Unreal Engine launcher > Open the library tab> Select the desired project.



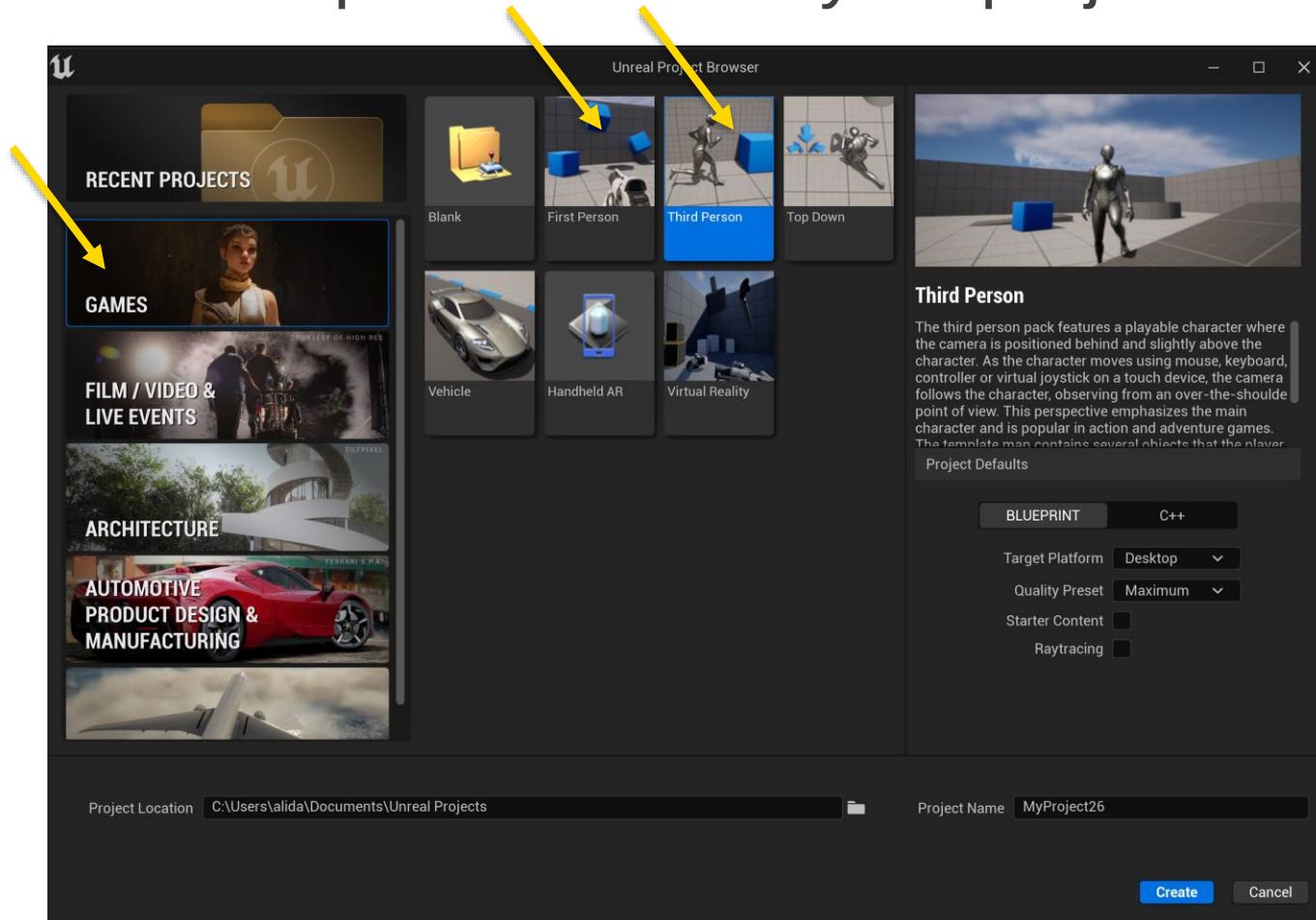
Create a new project

1. Select your project type

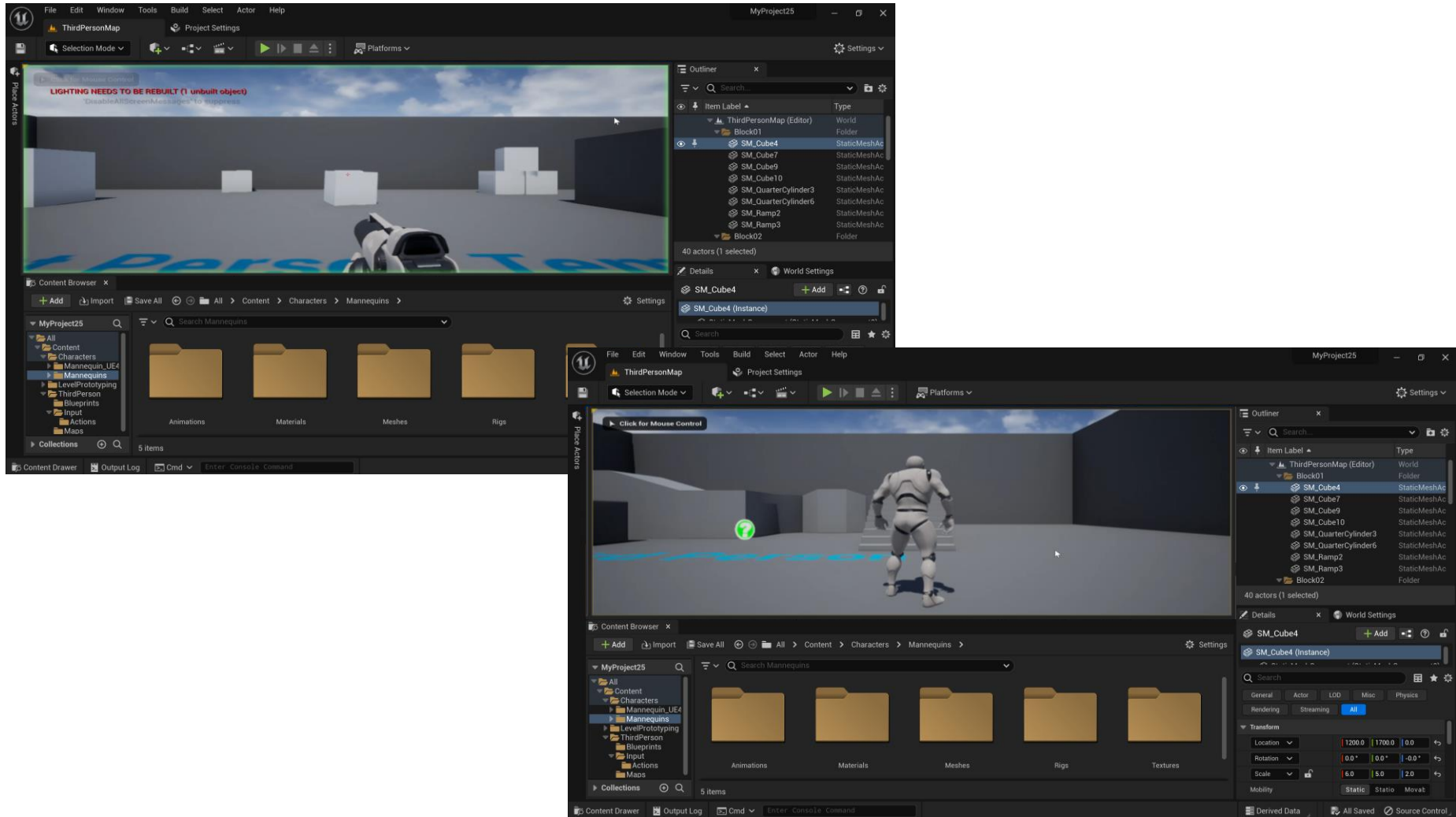


Create a new game project

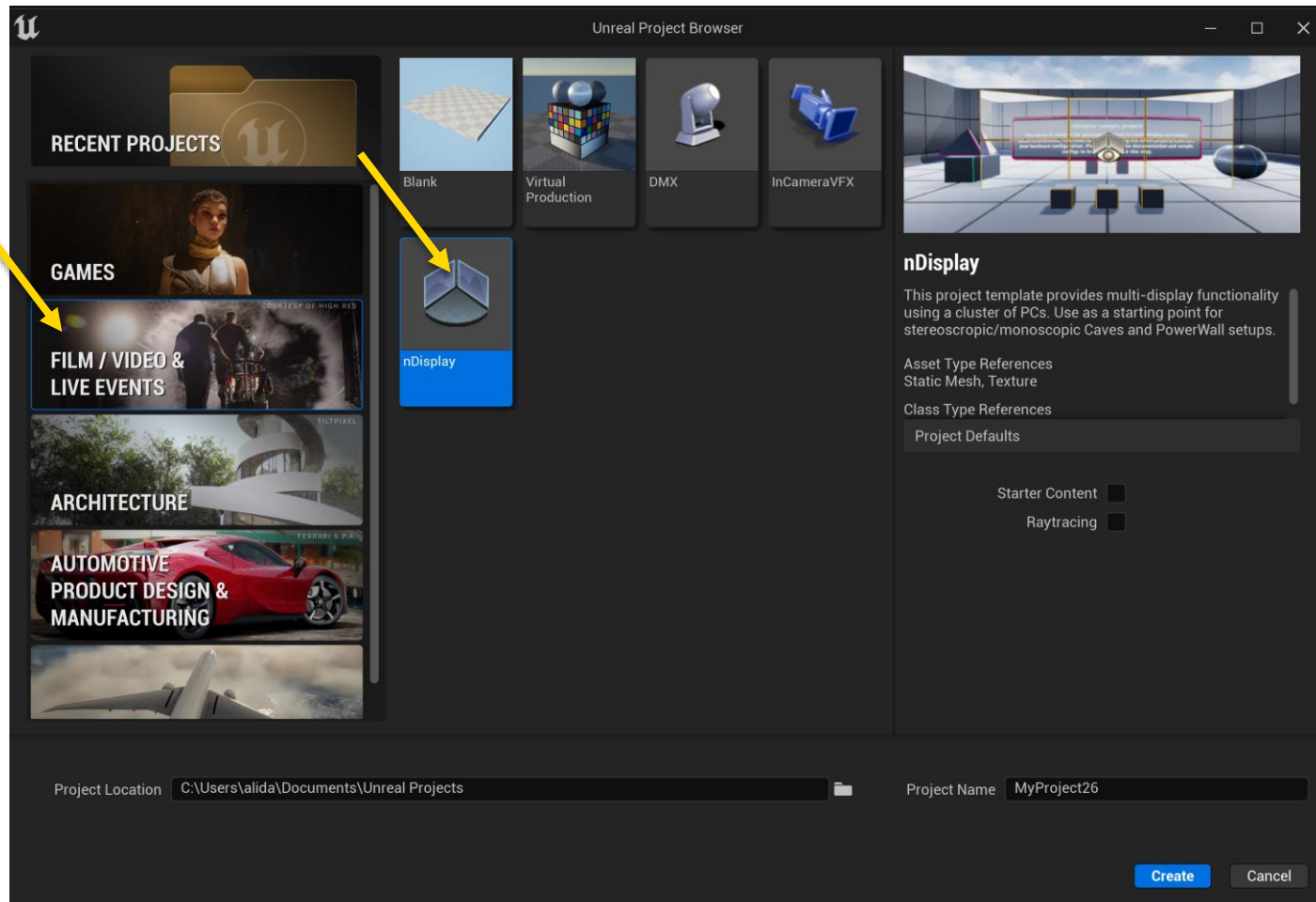
2. Select a template that suits your project



First person and third person templates



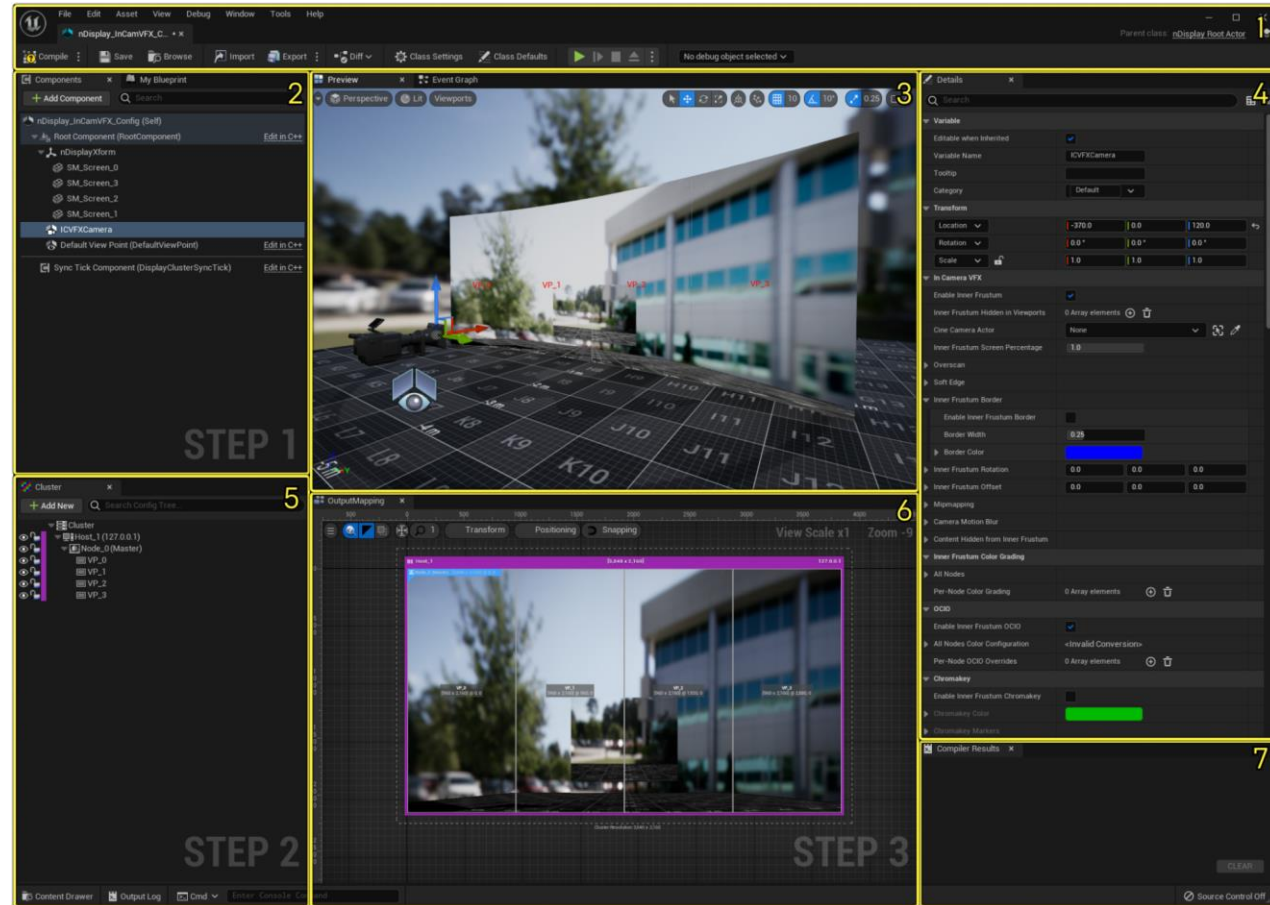
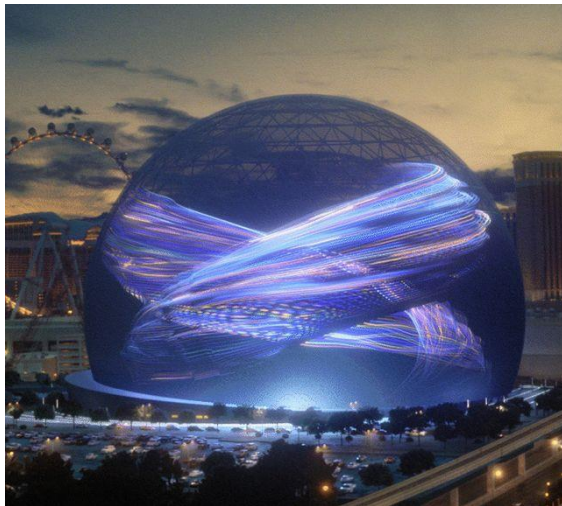
Create a new film project



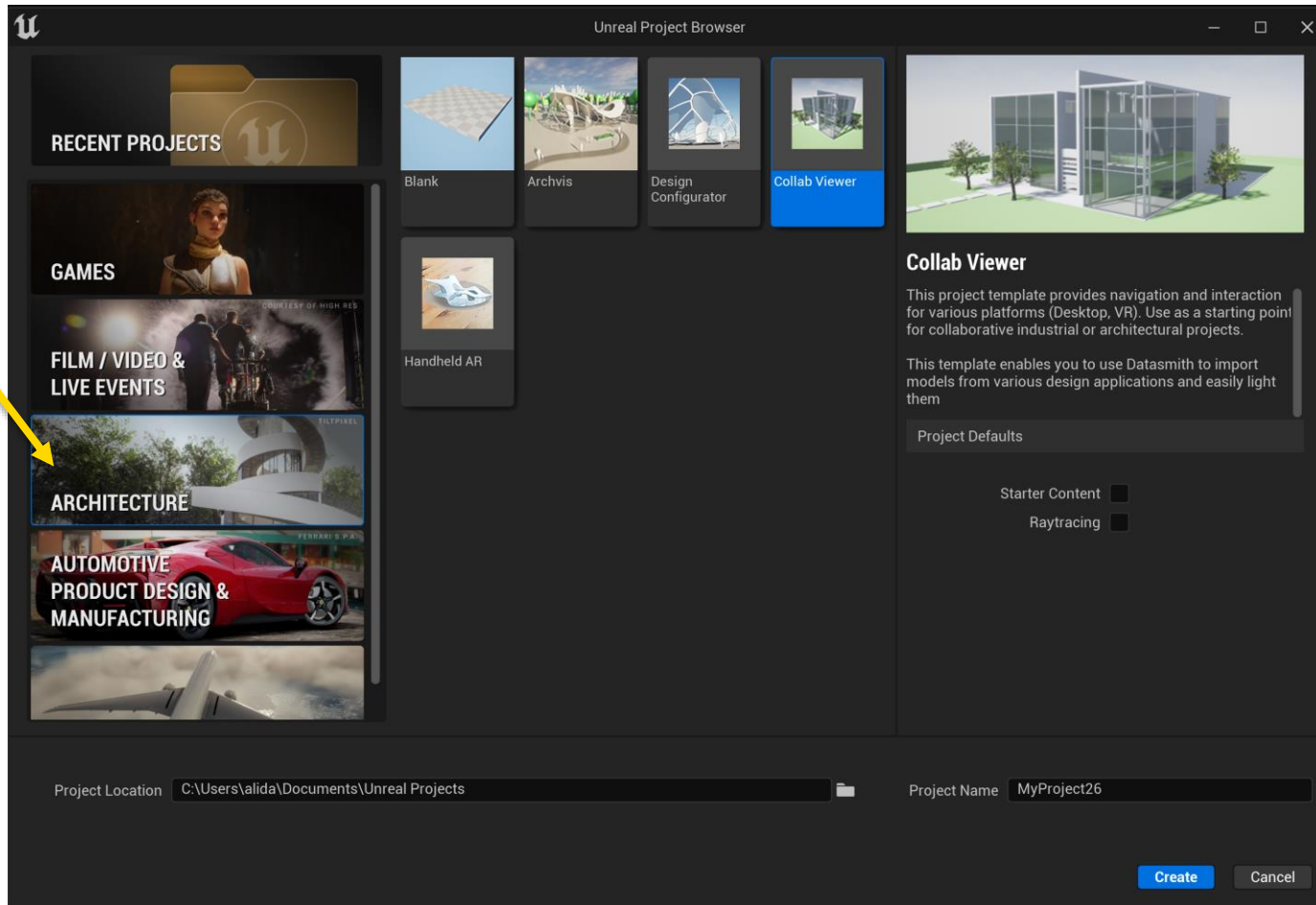
Create a new film project

nDisplay template is used for creating complex arrangements of physical displays for live performances.

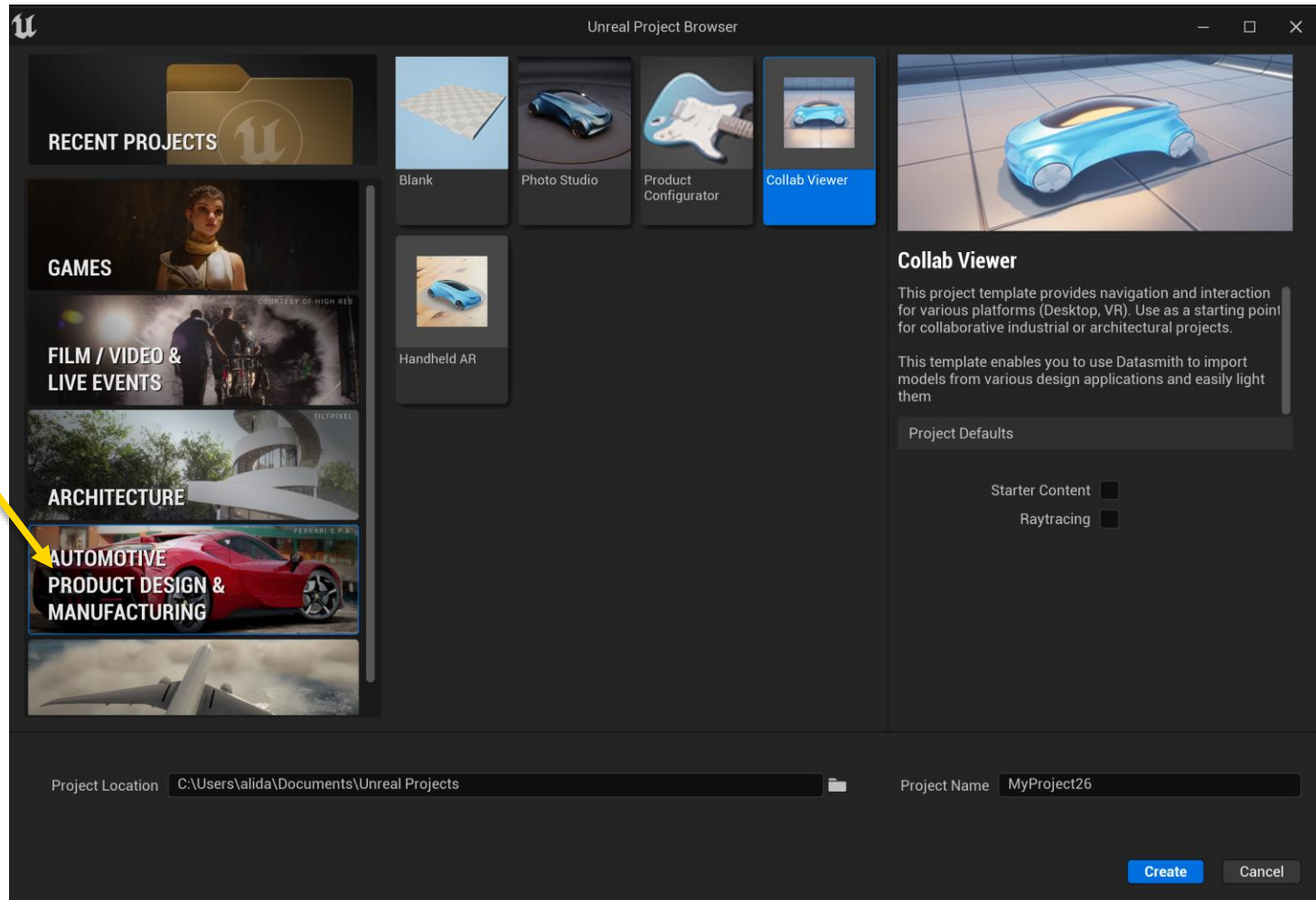
MSG Sphere



Create a new architecture project



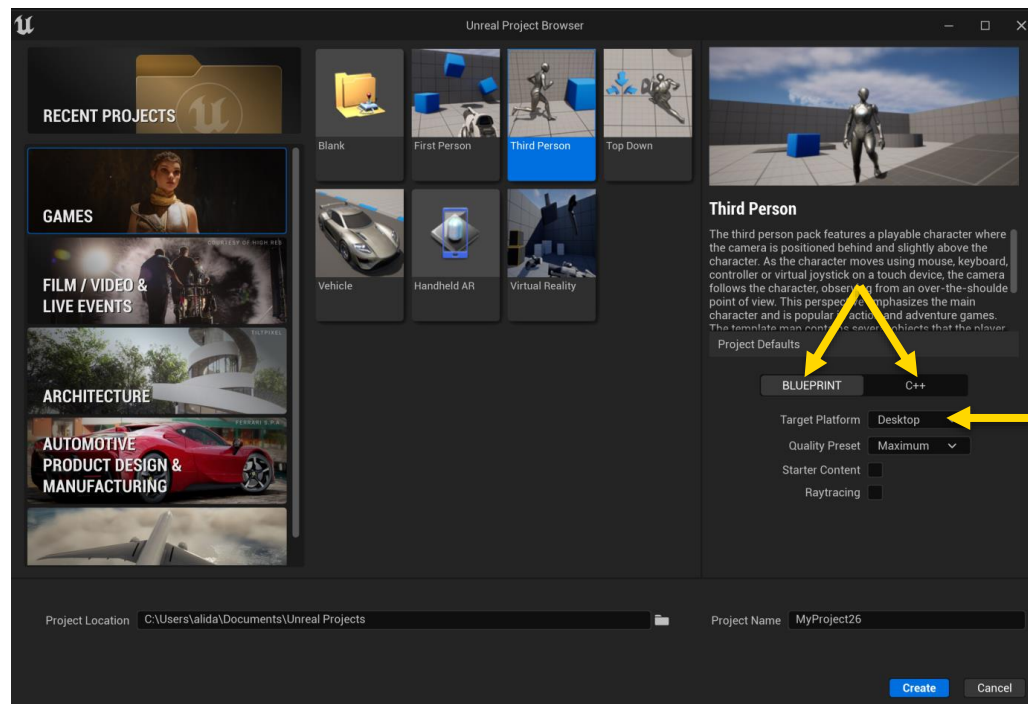
Create a new engineering project



Create a new game project

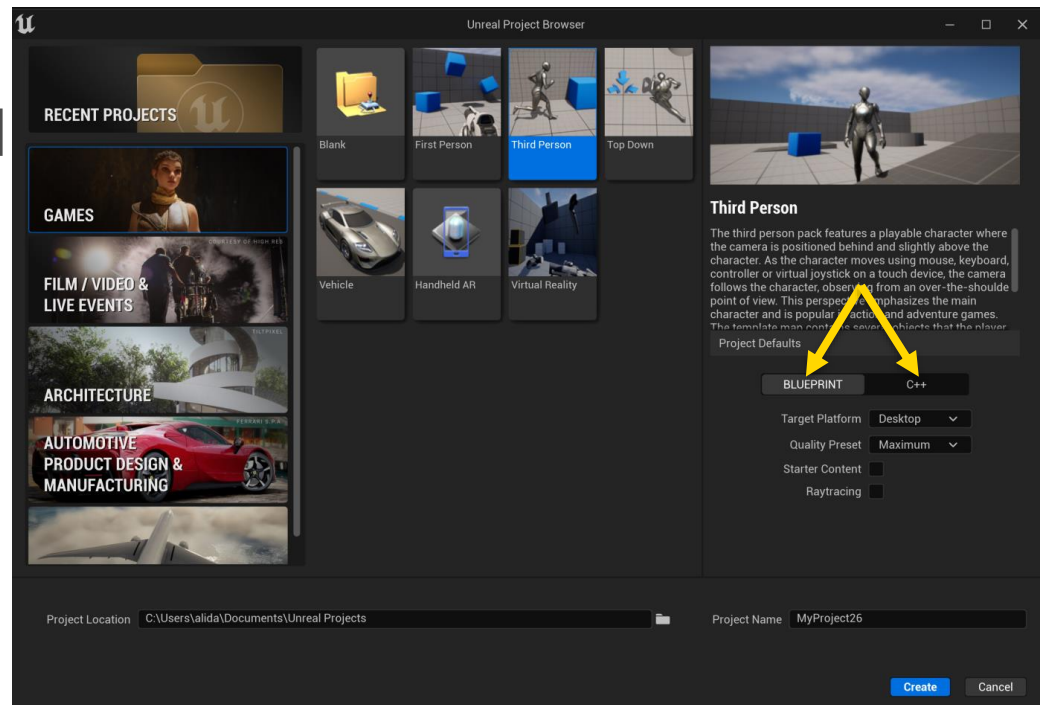
3. Select Type of coding (C++ or blueprint visual scripting)

4. Select desktop or mobile



C++ versus Blueprint visual scripting

- **Blueprints** is a visual scripting programming language within Unreal. Visual scripting is coding via a drag-and-drop method, connecting graphics rather than text.
- **C++** is a text-based programming language. It's low-level, yet versatile.

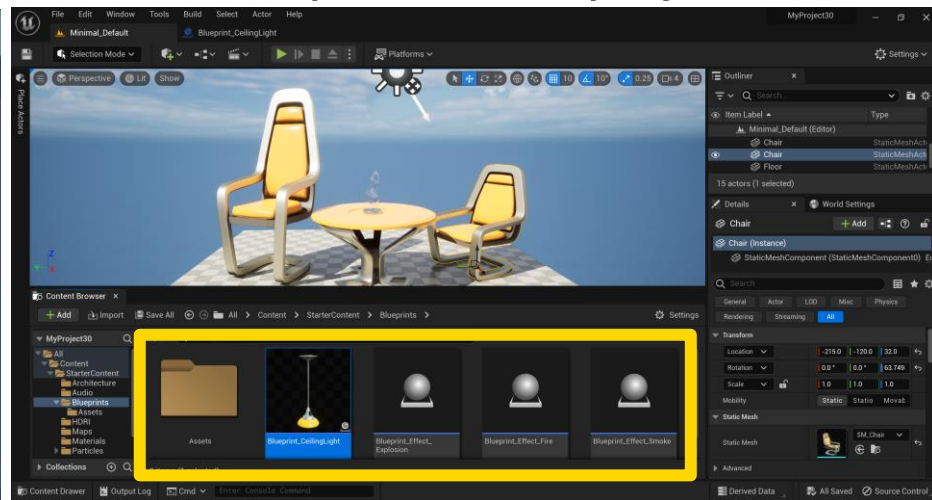
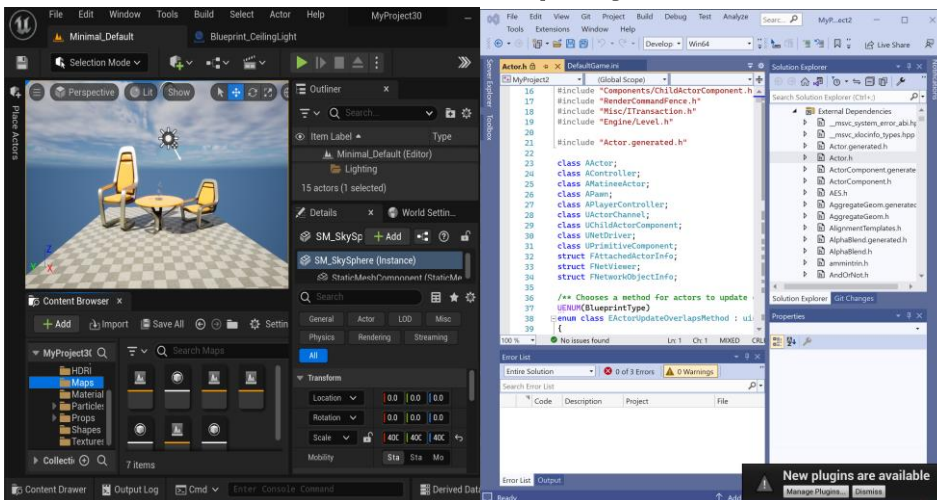


Can you add C++ code to a project that is created based on blueprints

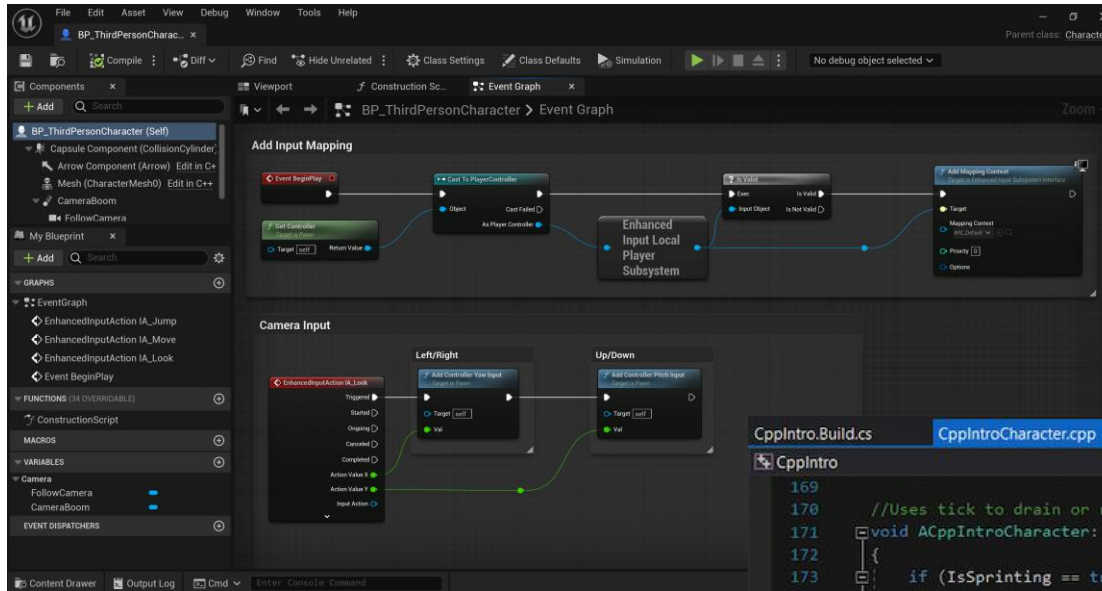
- Yes.
- The only difference is the template code that cannot be changed.
- In a Blueprint project, the base character and game type are implemented in Blueprint,
- In a C++ project, the base character and game type are implemented in C++.

C++ based project

Blueprint based project



C++ versus Blueprint visual scripting



```
CppIntro.Build.cs  CppIntroCharacter.cpp  CppIntroCharacter.h
CppIntro
169
170 //Uses tick to drain or regen stamina based on if the player is sprinting or not
171 void ACppIntroCharacter::Tick(float DeltaTime)
172 {
173     if (IsSprinting == true)
174     {
175         if (CurrentStamina > 0)
176         {
177             CurrentStamina -= DeltaTime;
178         }
179         else
180         {
181             SprintStop();
182         }
183     }
184
185     if (IsSprinting == false)
186     {
187         if (CurrentStamina < MaxStamina)
188         {
189             CurrentStamina += DeltaTime;
190         }
191     }
192 }
```

Blueprint visual scripting

Advantages

- Quicker to create a new game mechanic because of the ability to take advantage of pre-constructed nodes.
- It's much faster to get a prototype off the ground, and to get your game in a playable state.
- There is no need to be an expert programmer to use it.

Disadvantages

- Some very advanced engine features are limited or unavailable.
- As a higher-level language, in general Blueprint will run slower than C++.
- Large graphs with dozens of nodes can become unwieldy and difficult to read.

C++ scripting

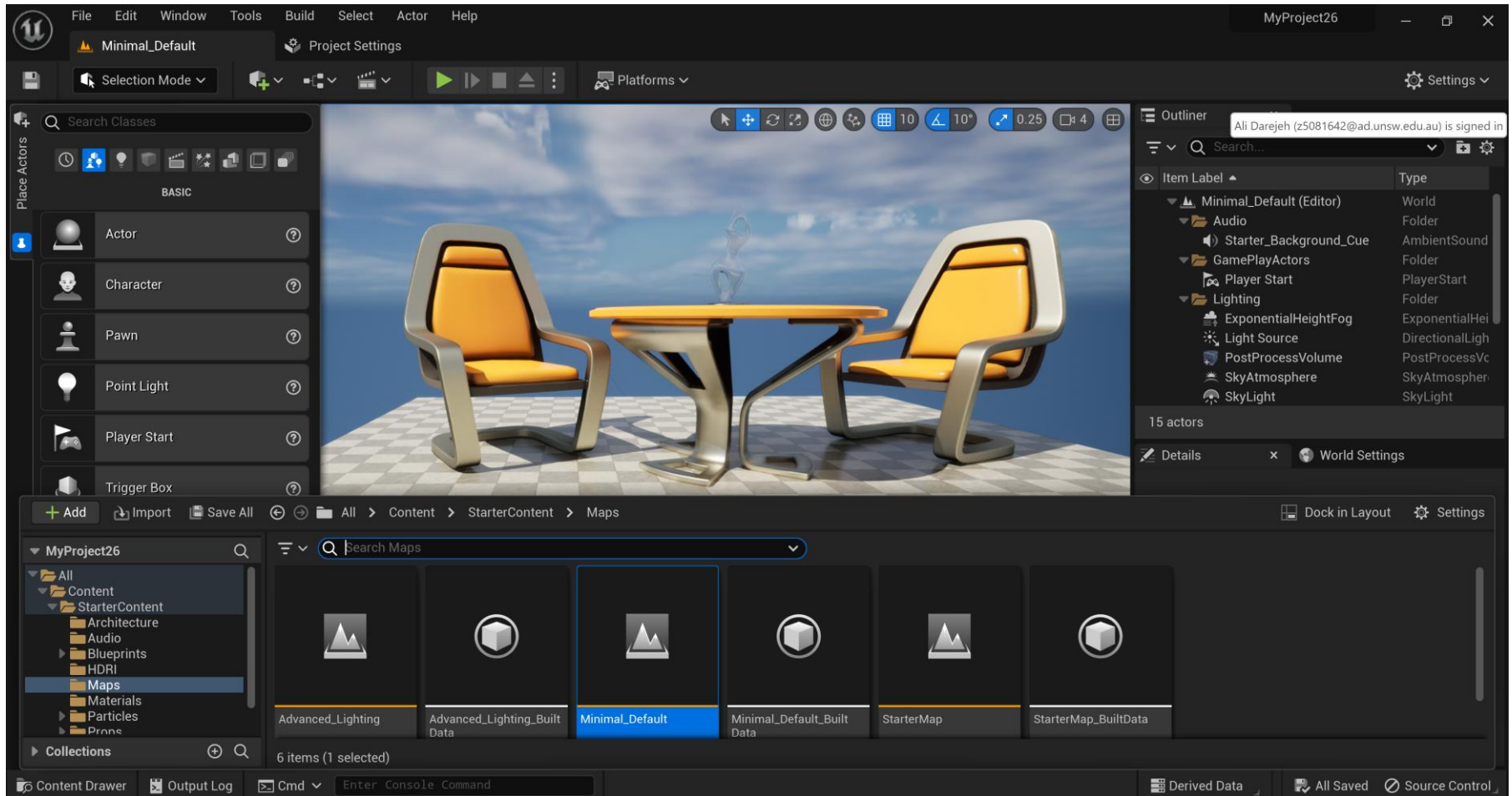
Advantages

- All of Unreal Engine's features are available right from the start.
- Those familiar with C++ will be able to dive into the engine's source code.
- Will generally execute faster, especially for low level operations.

Disadvantages

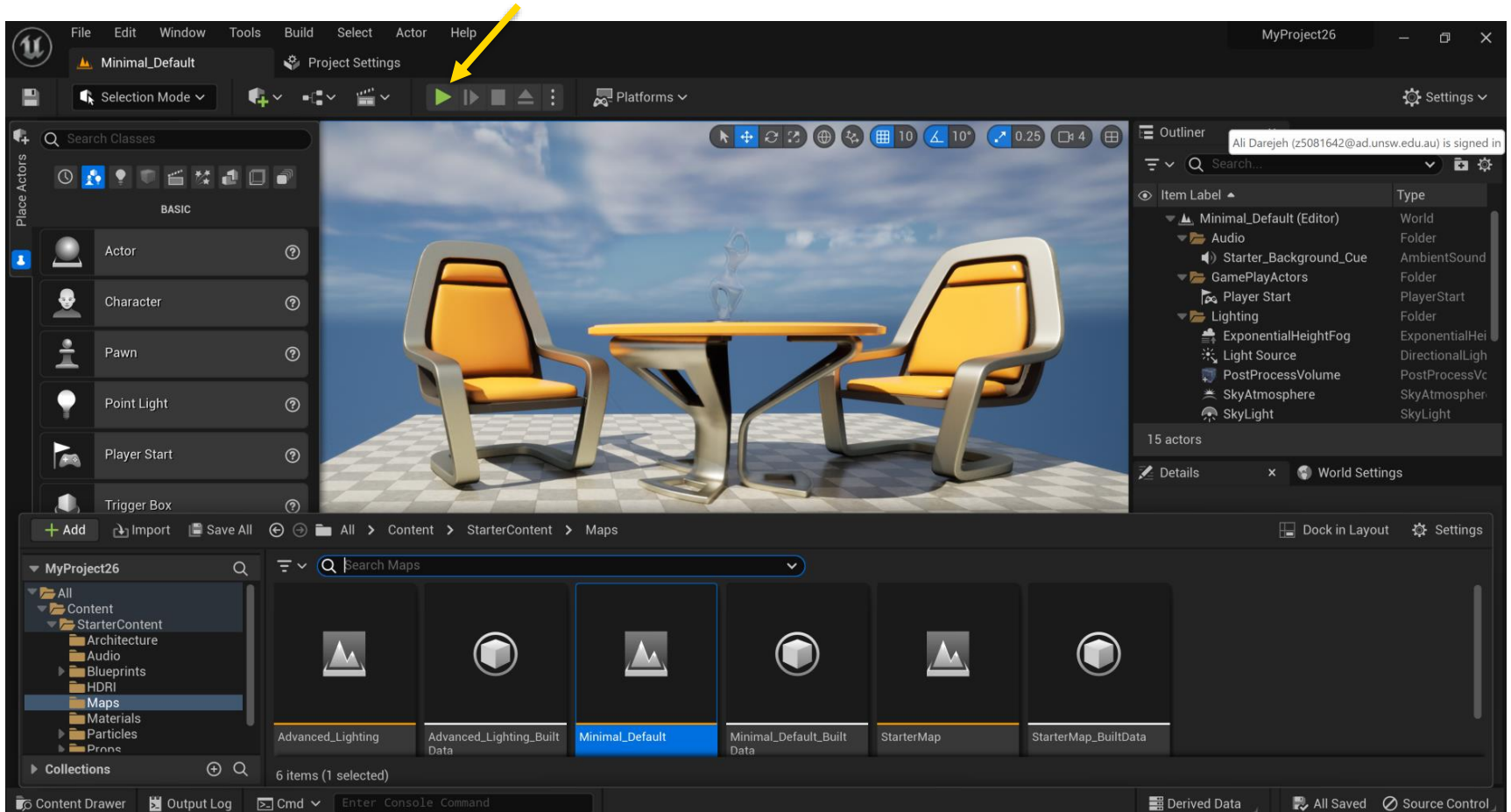
- There is a greater barrier of entry.
- It will take more work early in production to get your game up and running.
- There is less of a safety net, making it easier to crash your game.

Unreal Engine environment



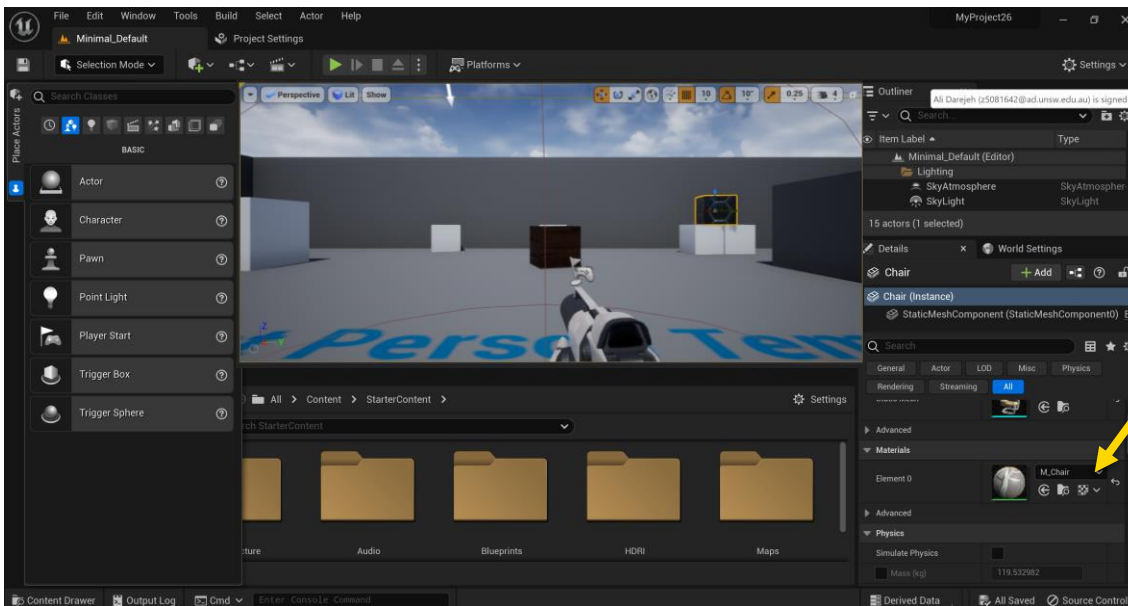
Unreal Engine environment

In order to run your game click the run button

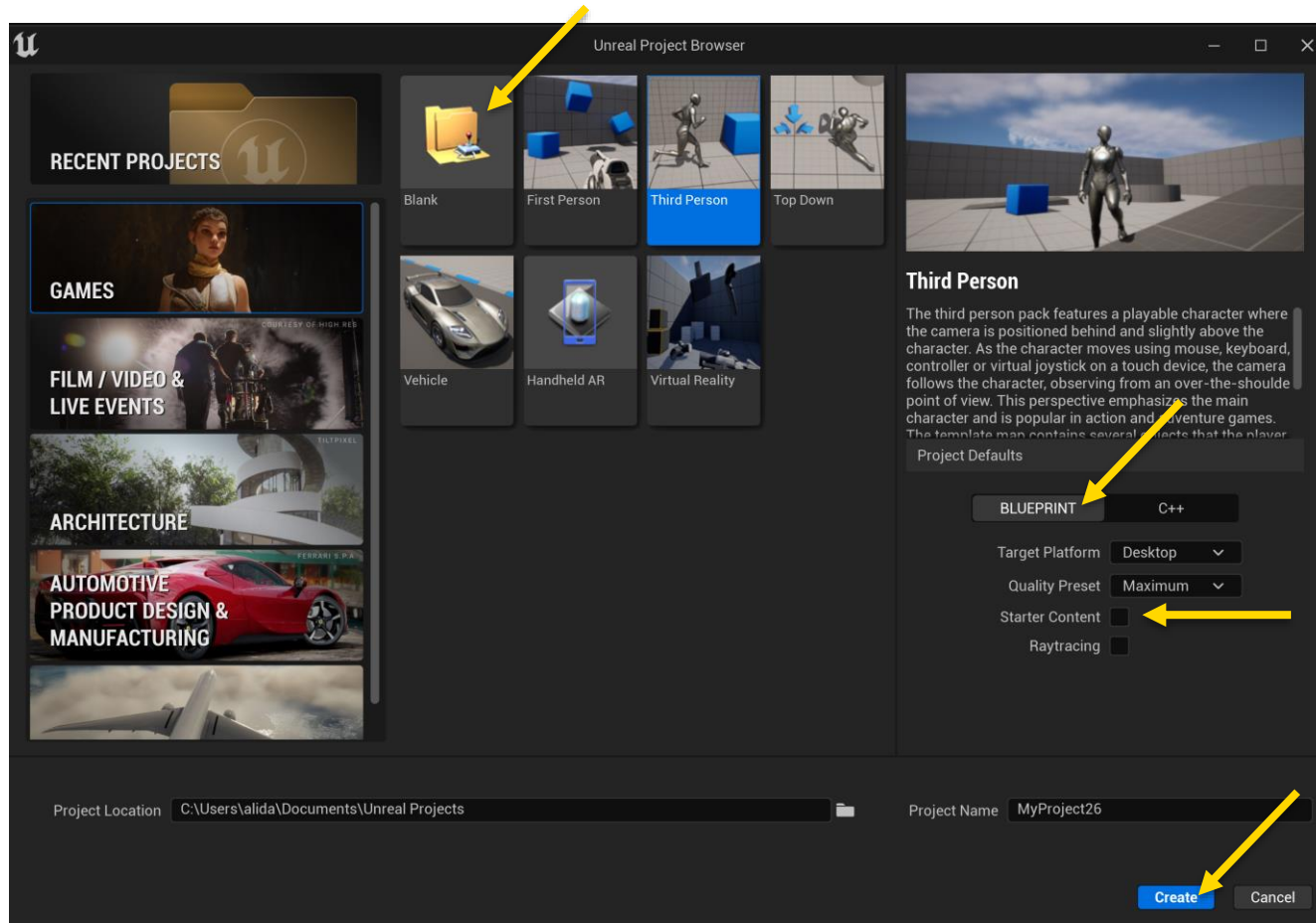


Lecture activity 1

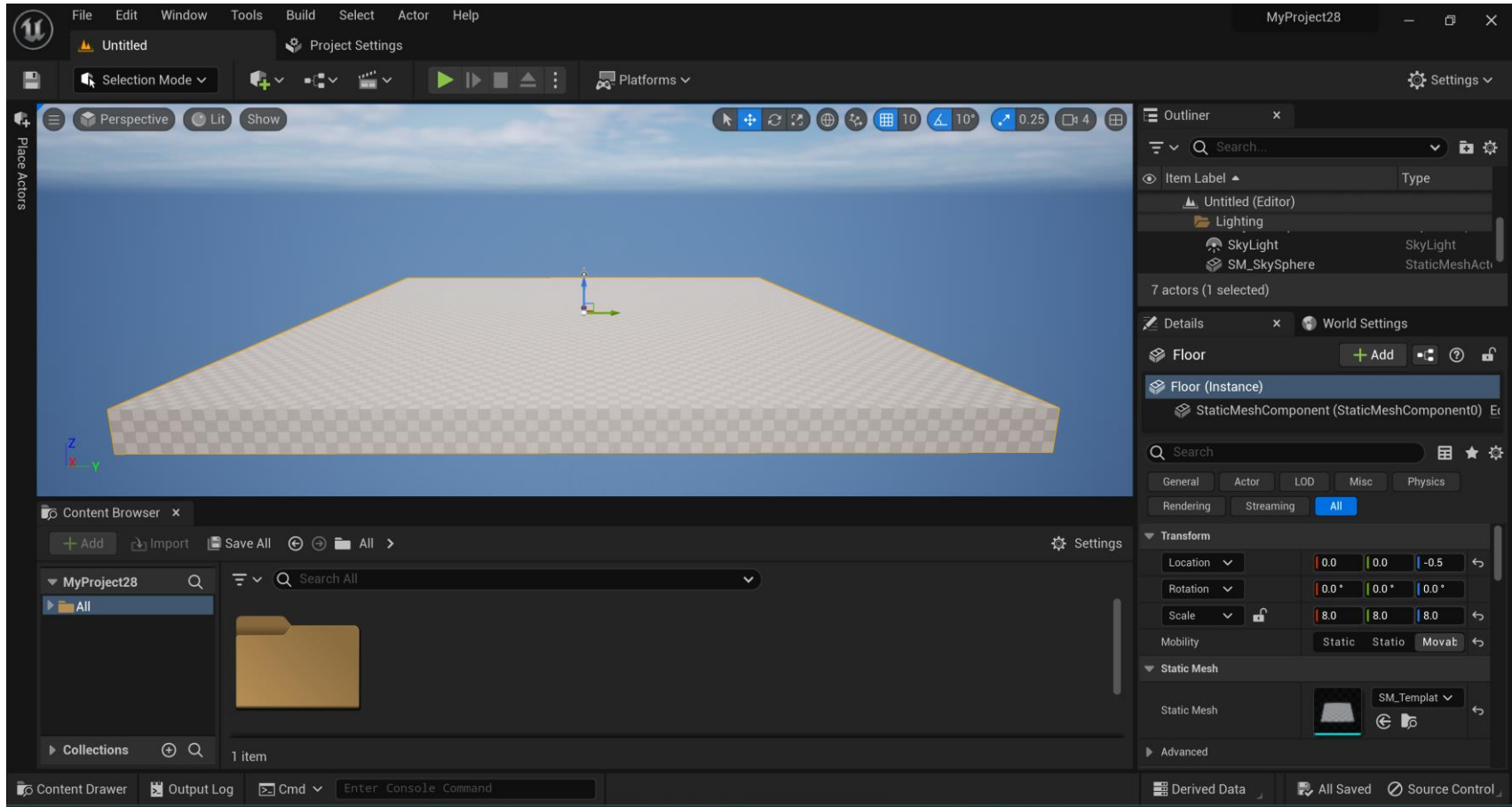
- Create a first-person shooter game project
- Change the material of the boxes to wood or metal
- Create a cone that falls off from the sky (You should enable physics and gravity)



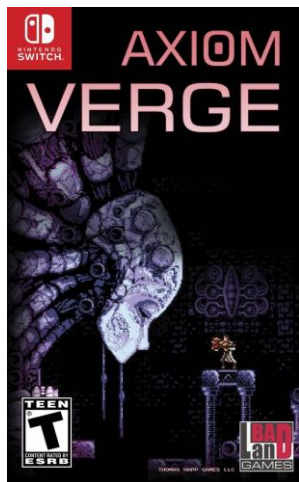
Create a blank project



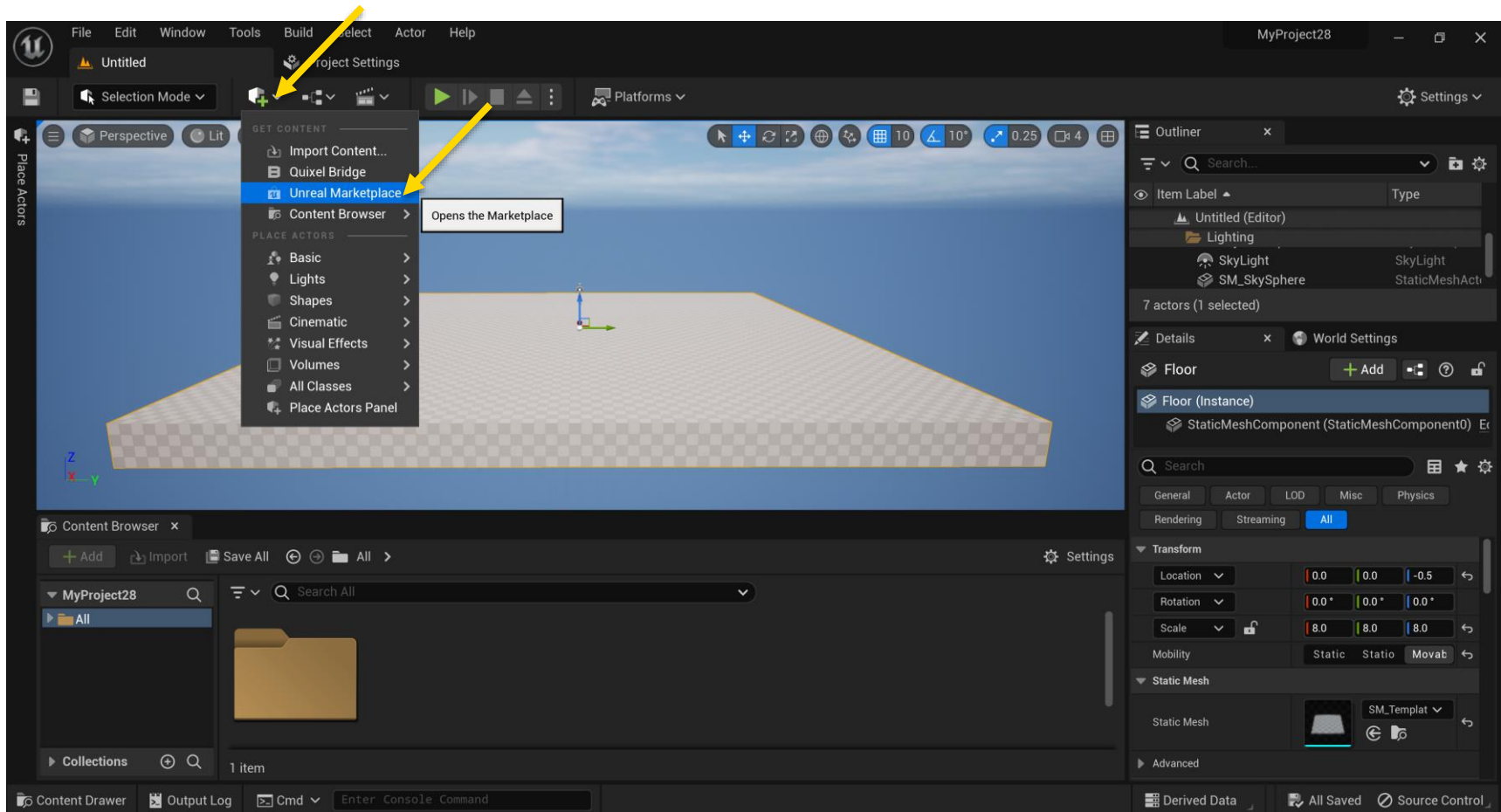
Create a blank project



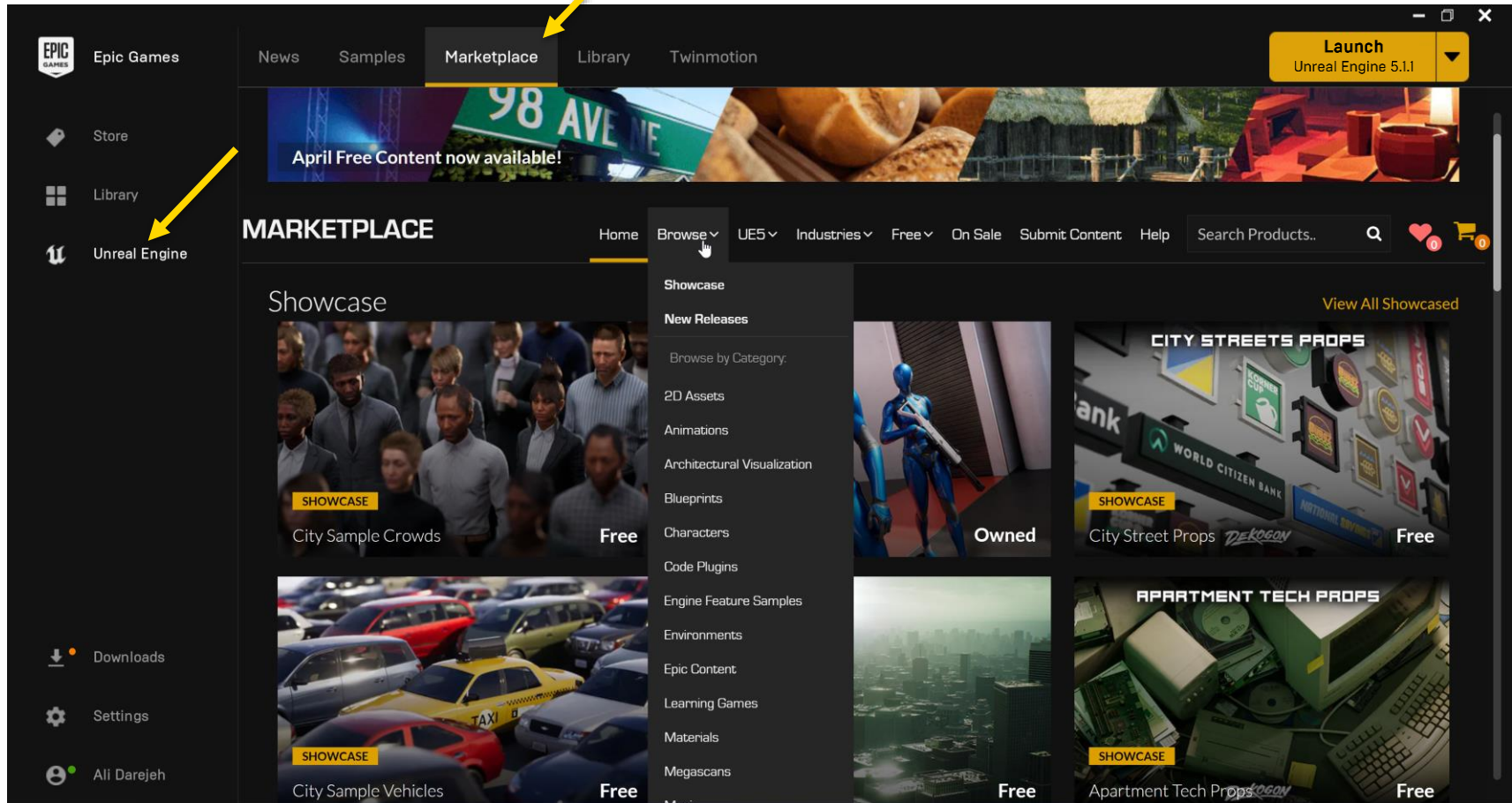
The games that are developed by 1-3 developers



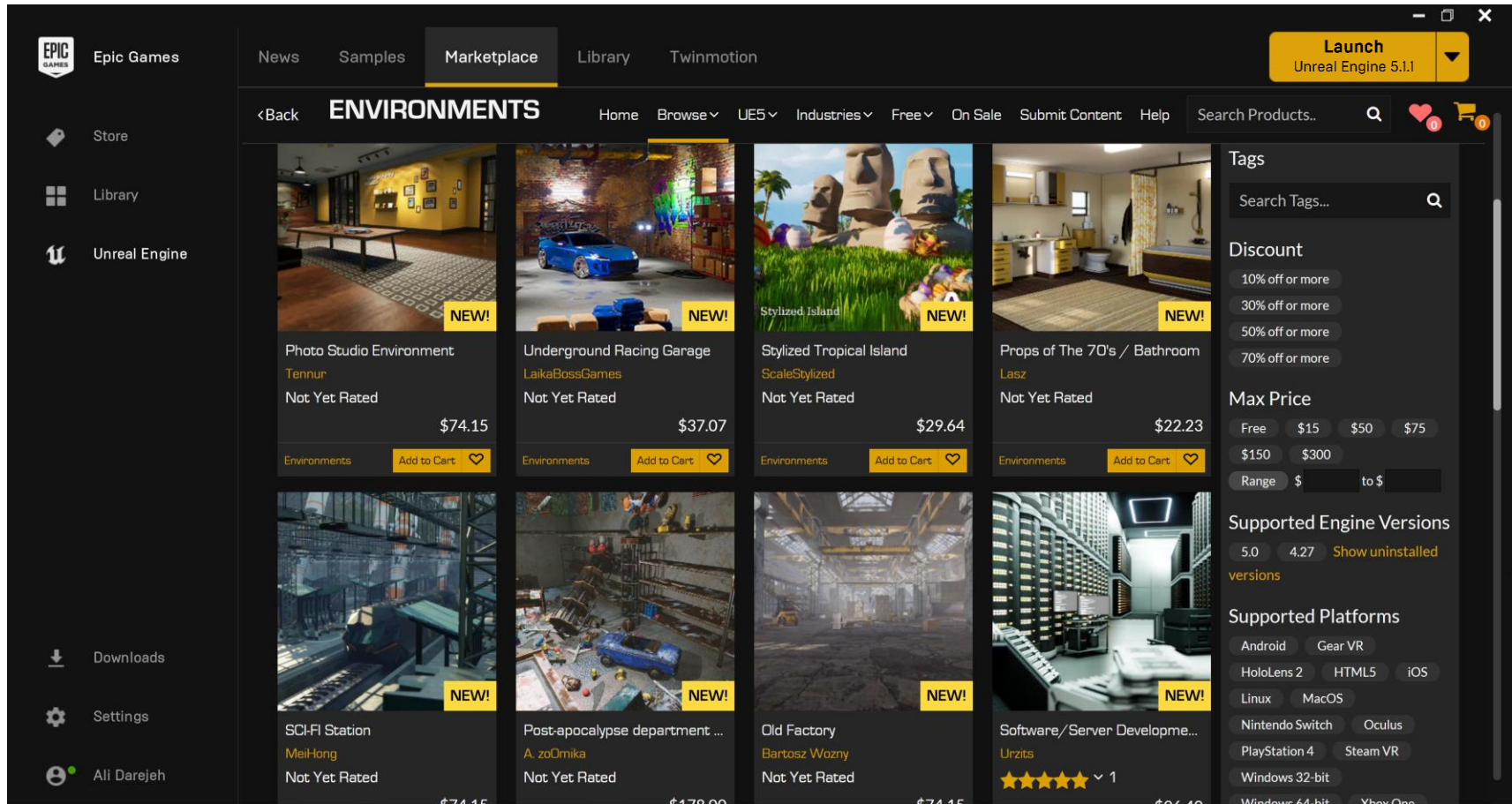
Unreal Engine asset store



A horizontal bar with a yellow arrow pointing to the right end.



Unreal Engine asset store



Unreal Engine asset store

The screenshot displays the Unreal Engine Marketplace interface. On the left is a dark sidebar with the Epic Games logo and navigation links: Store, Library, Unreal Engine, Downloads, Settings, and a user profile for Ali Darejeh. The top navigation bar includes links for News, Samples, Marketplace (which is highlighted), Library, and Twinmotion. A yellow 'Launch Unreal Engine 5.1.1' button is in the top right. Below the navigation is a banner for 'April Free Content now available!' featuring a collage of assets including a '98 AVENUE' street sign, bread, and a thatched hut. The main content area is titled 'CONTENT DETAIL' and shows the 'Apartment Tech Props' asset pack by Dekogon Studios. The asset is marked as 'Free' and has a 5-star rating from 7 reviews. A large image shows a detailed 3D rendering of a desk cluttered with various electronic components and peripherals. Below this image are four smaller thumbnail images. To the right of the main image, the asset's title 'Apartment Tech Props' is displayed, followed by the creator's name 'Dekogon Studios - Environments' and the release date 'Apr 2, 2022'. The rating and review information are shown below the title. A description states: 'A collection of apartment tech assets that can be used for games!'. At the bottom right, there are buttons for 'Free', 'OR', 'Add to Cart', and a heart icon. To the right of these buttons, a table lists the supported platforms (Windows, macOS, Linux, iOS, Android), supported engine versions (5.0), and download type (Asset Pack).

Epic Games Epic Games

News Samples Marketplace Library Twinmotion

Launch Unreal Engine 5.1.1

April Free Content now available!

<Back CONTENT DETAIL Home Browse UE5 Industries Free On Sale Submit Content Help Search Products..

Apartment Tech Props

Dekogon Studios - Environments - Apr 2, 2022

★★★★★ 7 7 reviews written | 1 of 1 question answered

A collection of apartment tech assets that can be used for games!

Free

OR

Add to Cart

Supported Platforms

Supported Engine Versions

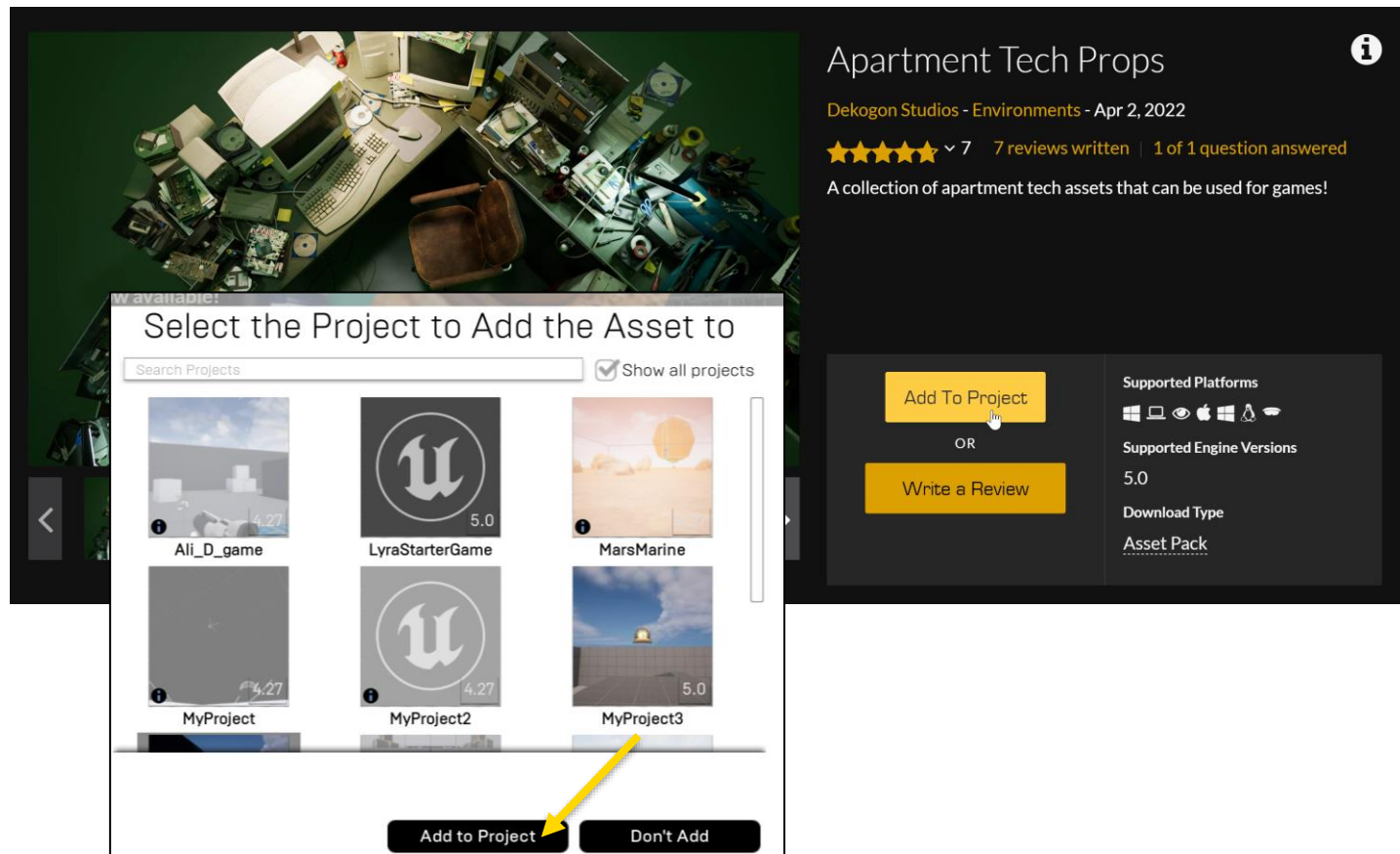
5.0

Download Type

Asset Pack

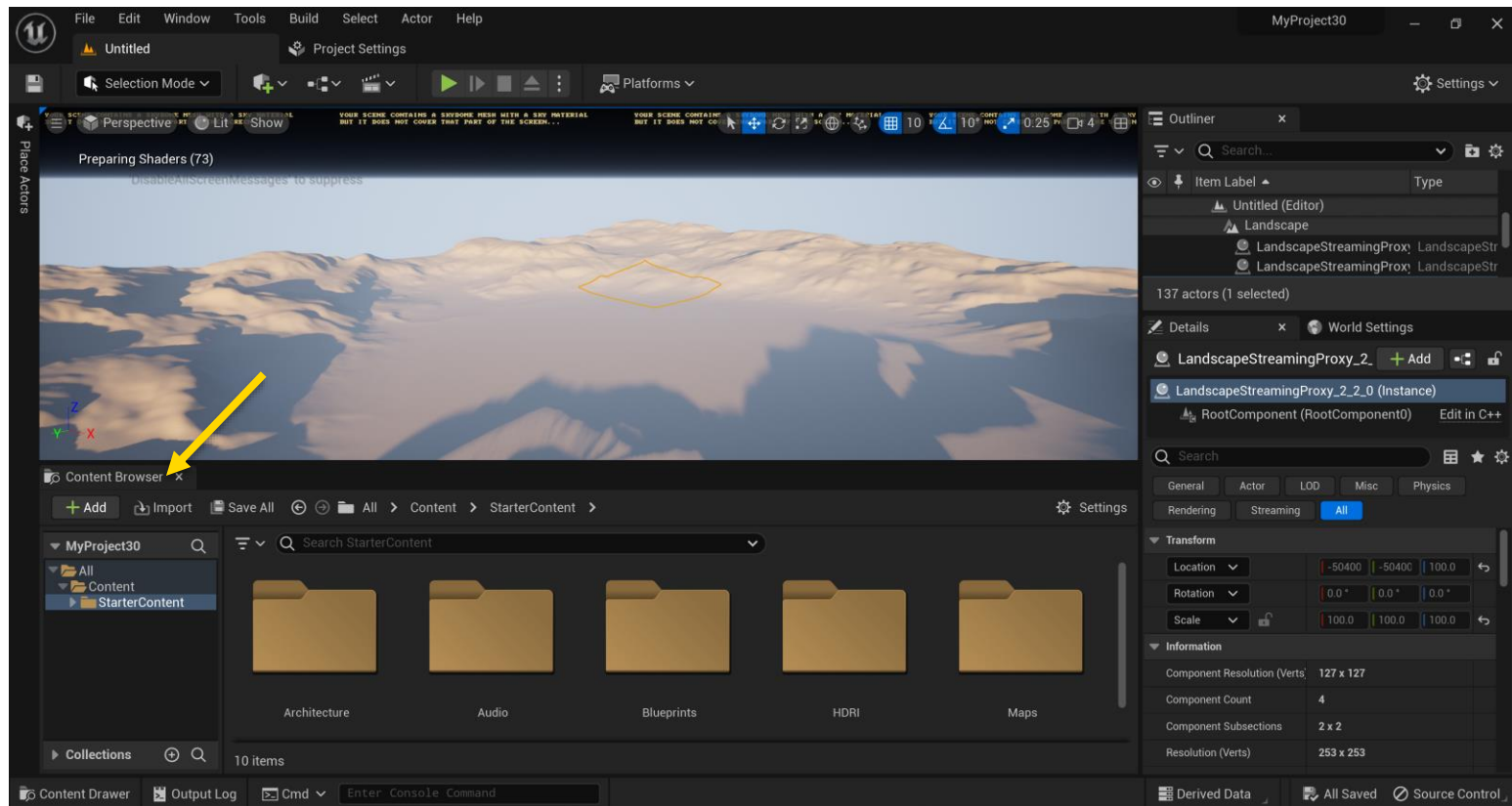
Unreal Engine asset store

After downloading an asset you should add it to your project by clicking and to project button



Unreal Engine asset store

After adding the assets, you can view them in the content browser of your project.



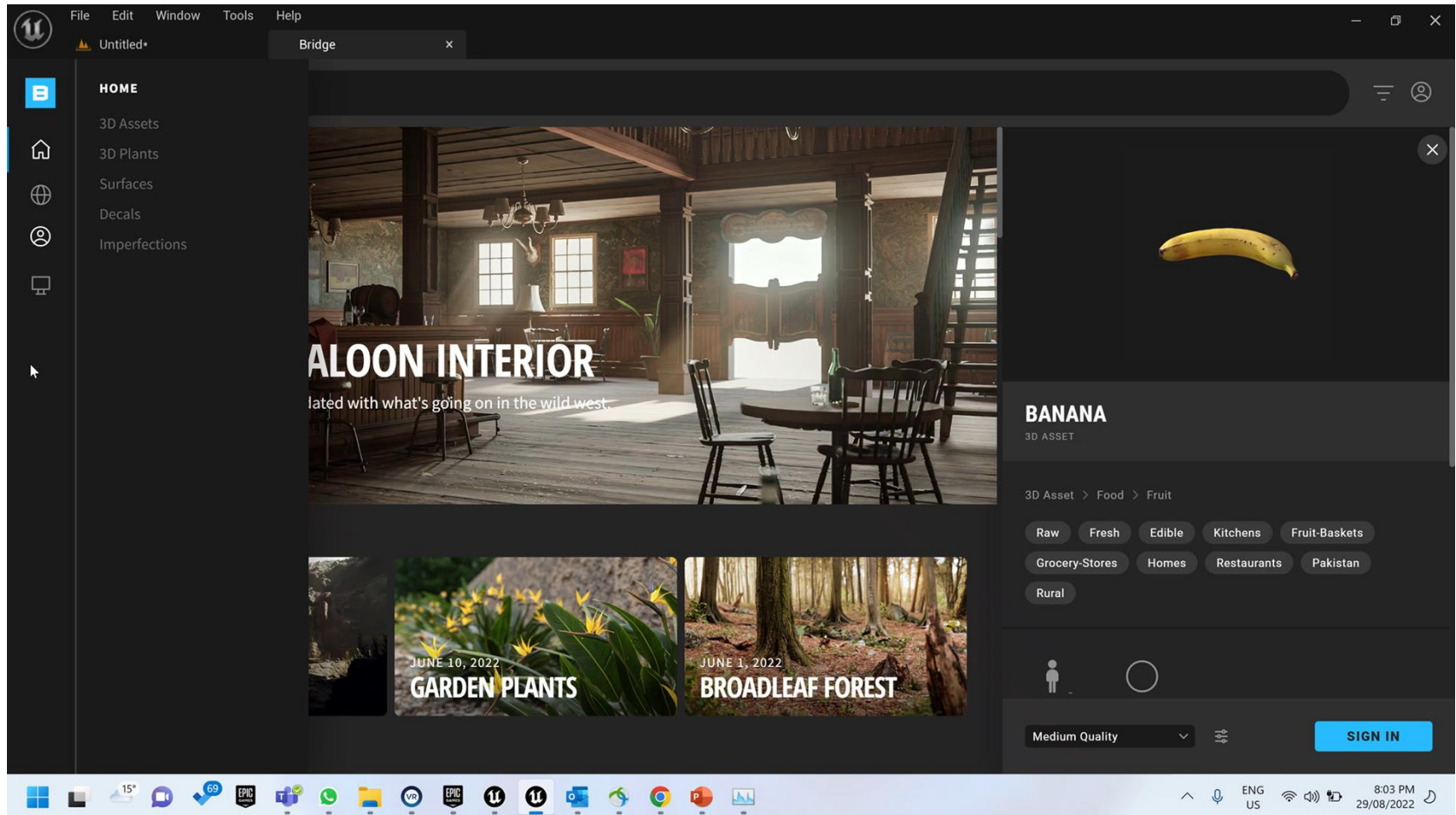
Lecture activity 2

- Go to Unreal Engine asset store find a free asset and add it into your level.



Unreal Engine asset store (Quixel Bridge)

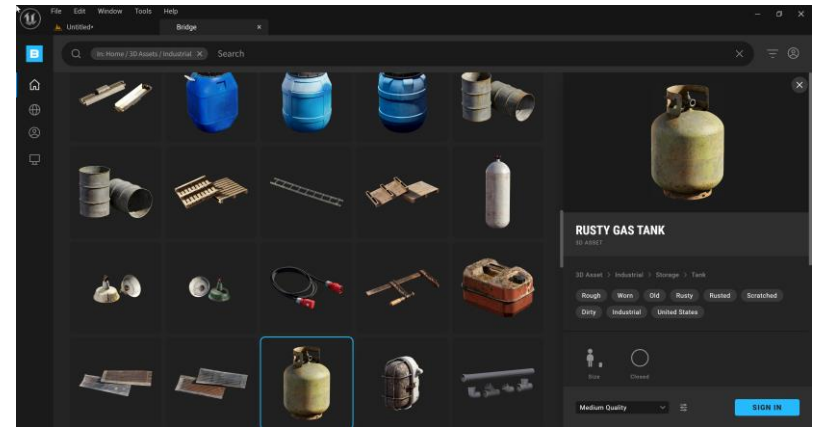
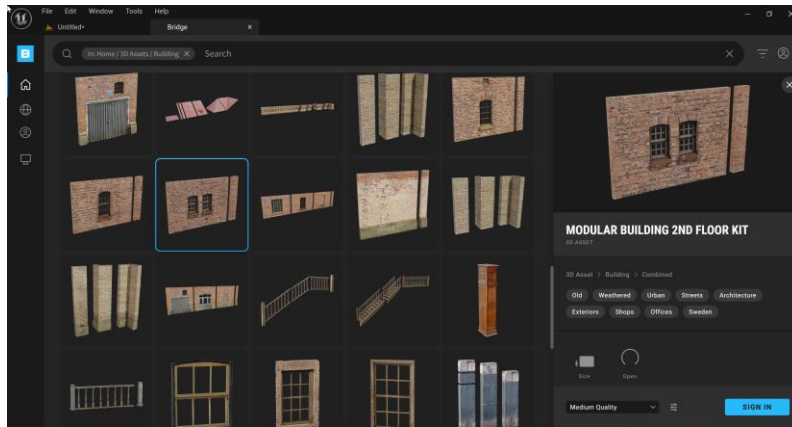
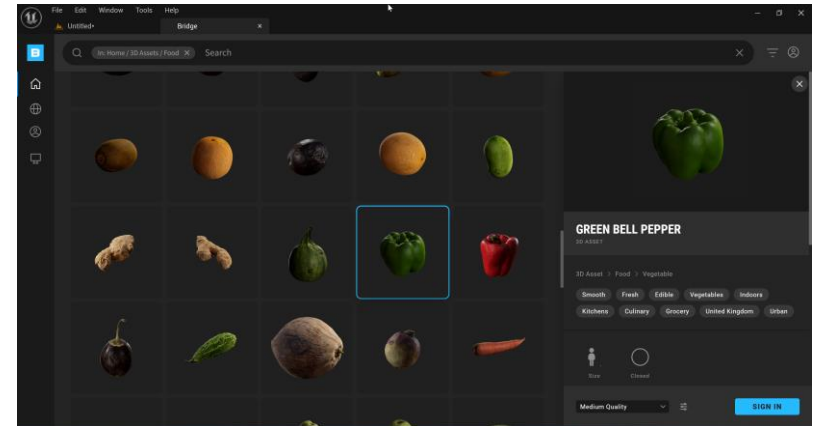
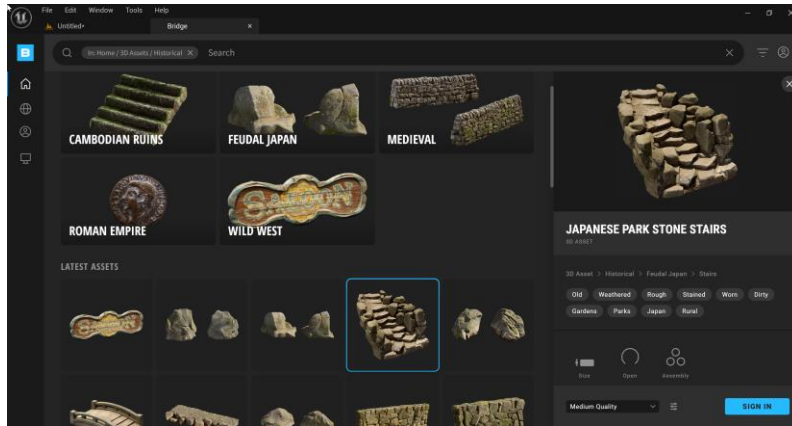
Quixel Bridge has over 16,000 photorealistic 3d scans



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

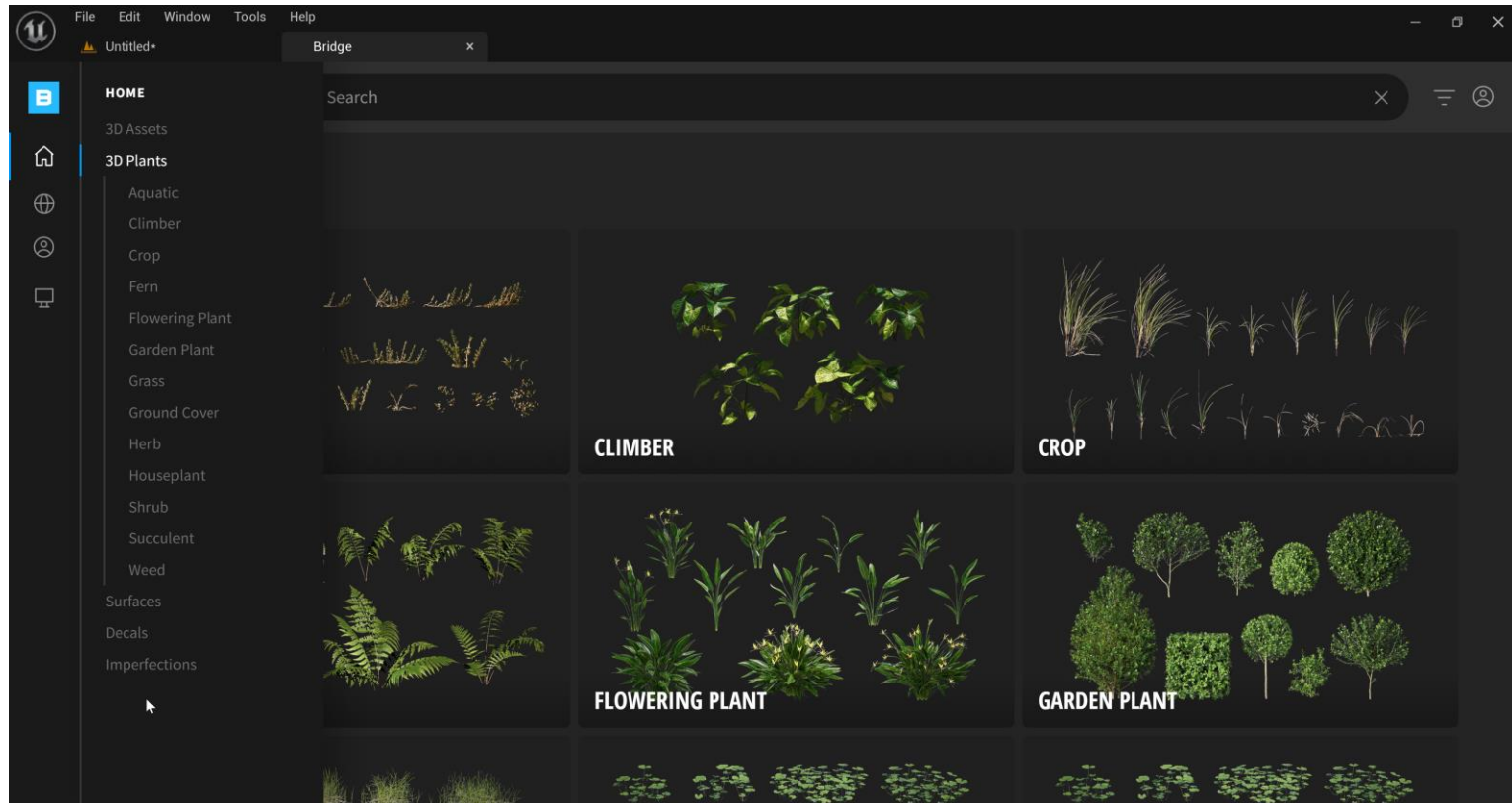
- Objects, foods, fruits



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

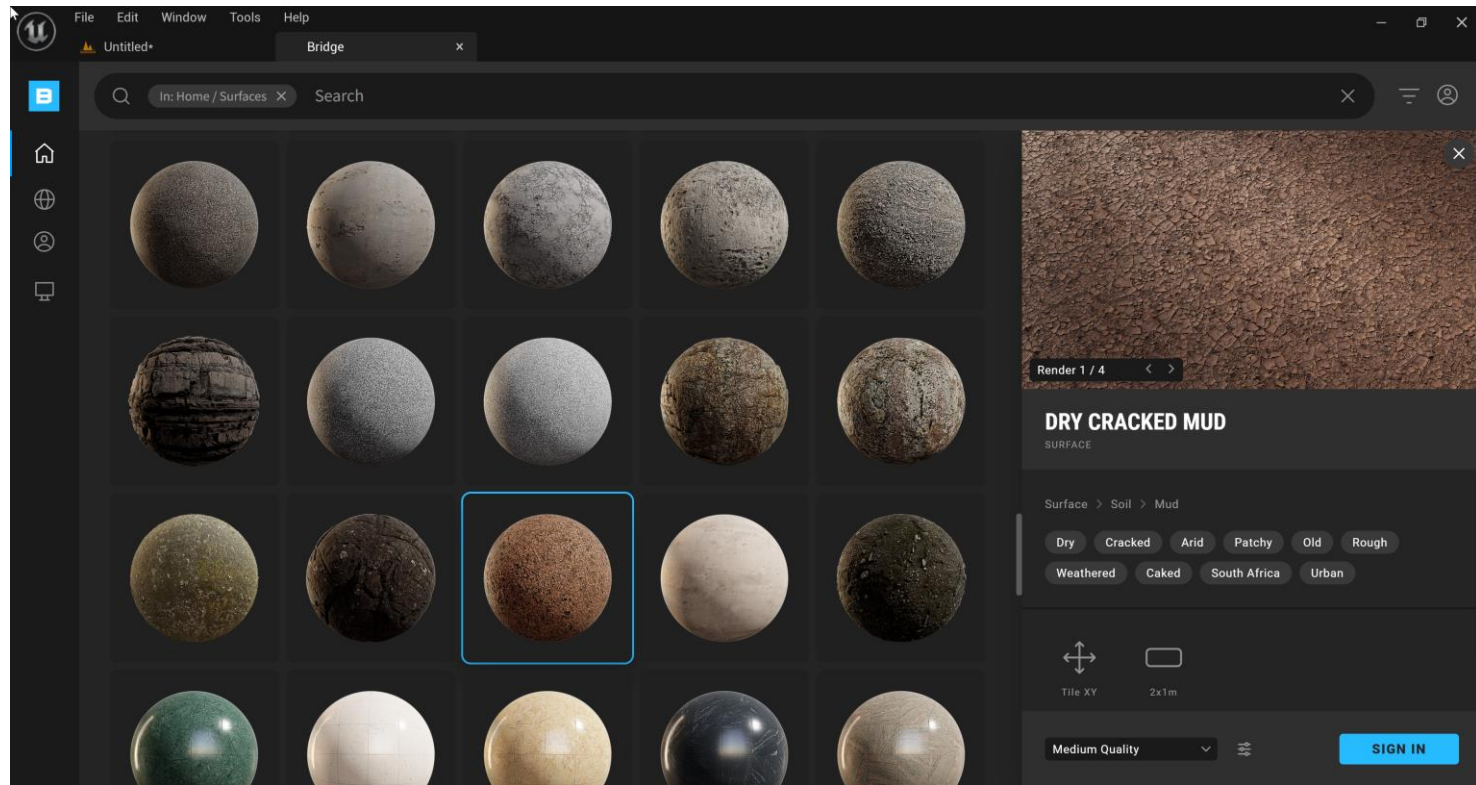
- Nature and plants



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

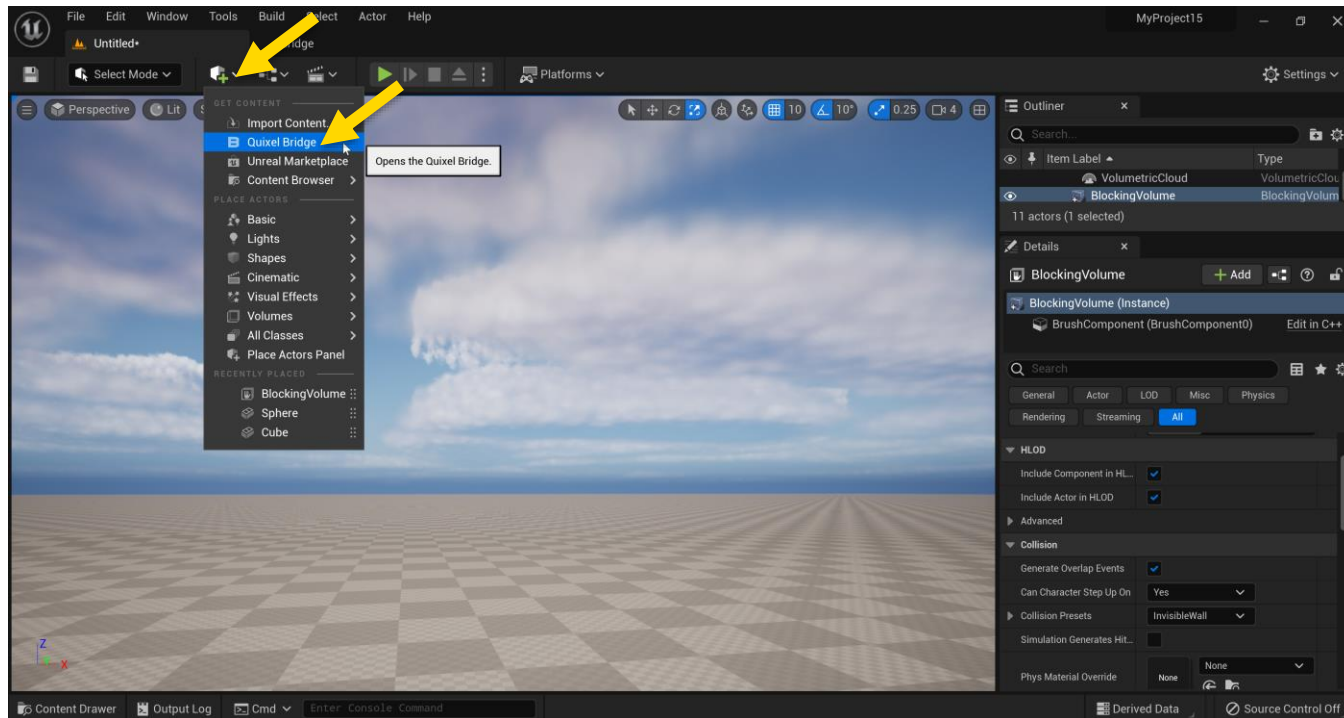
- Different materials that can be applied to objects.



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

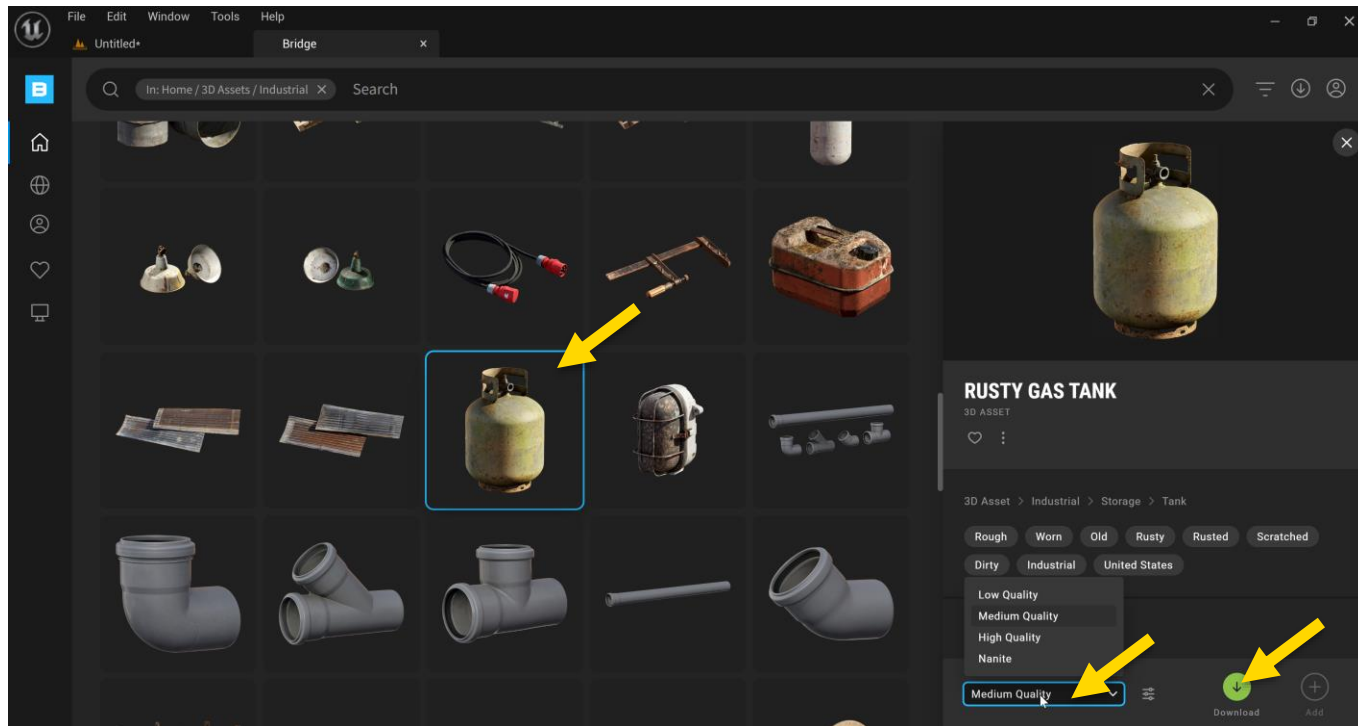
- Unreal Engine 5 > Click the **Add Object** button > Click Quixel Bridge.



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

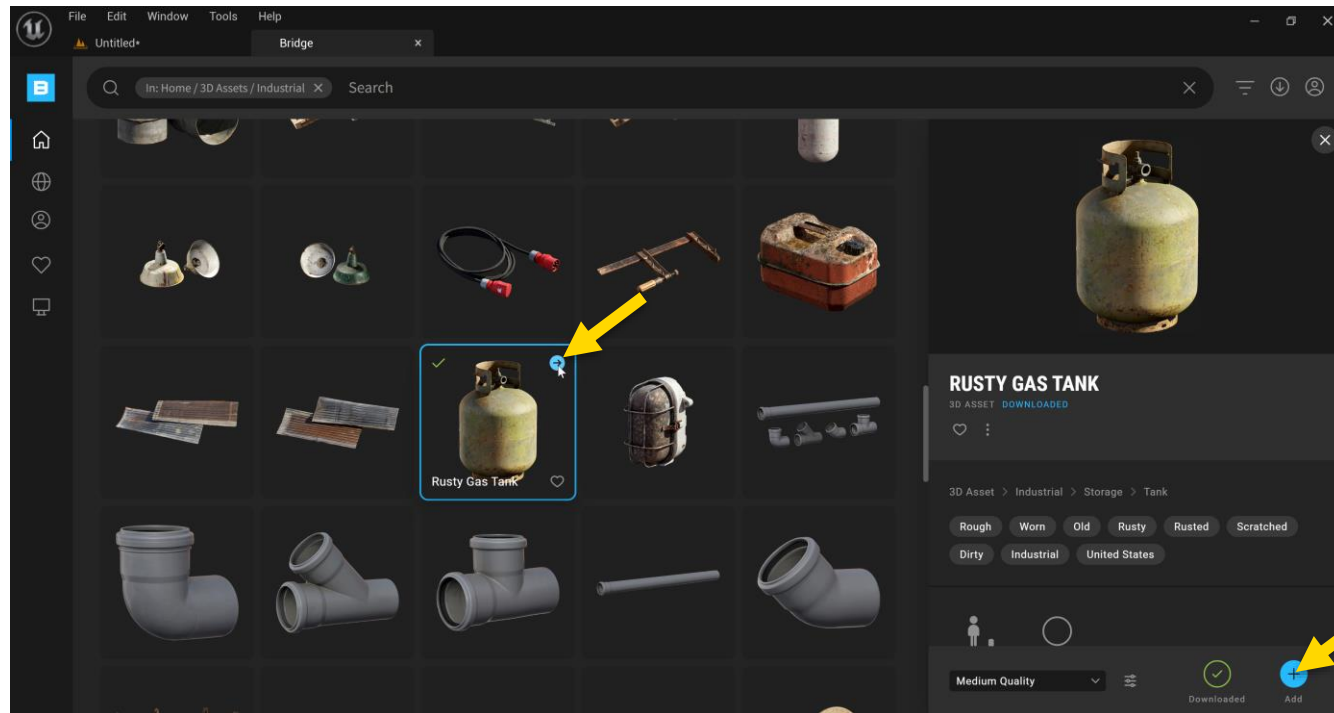
- Select the desired object > Select the quality > Click the Download button.



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

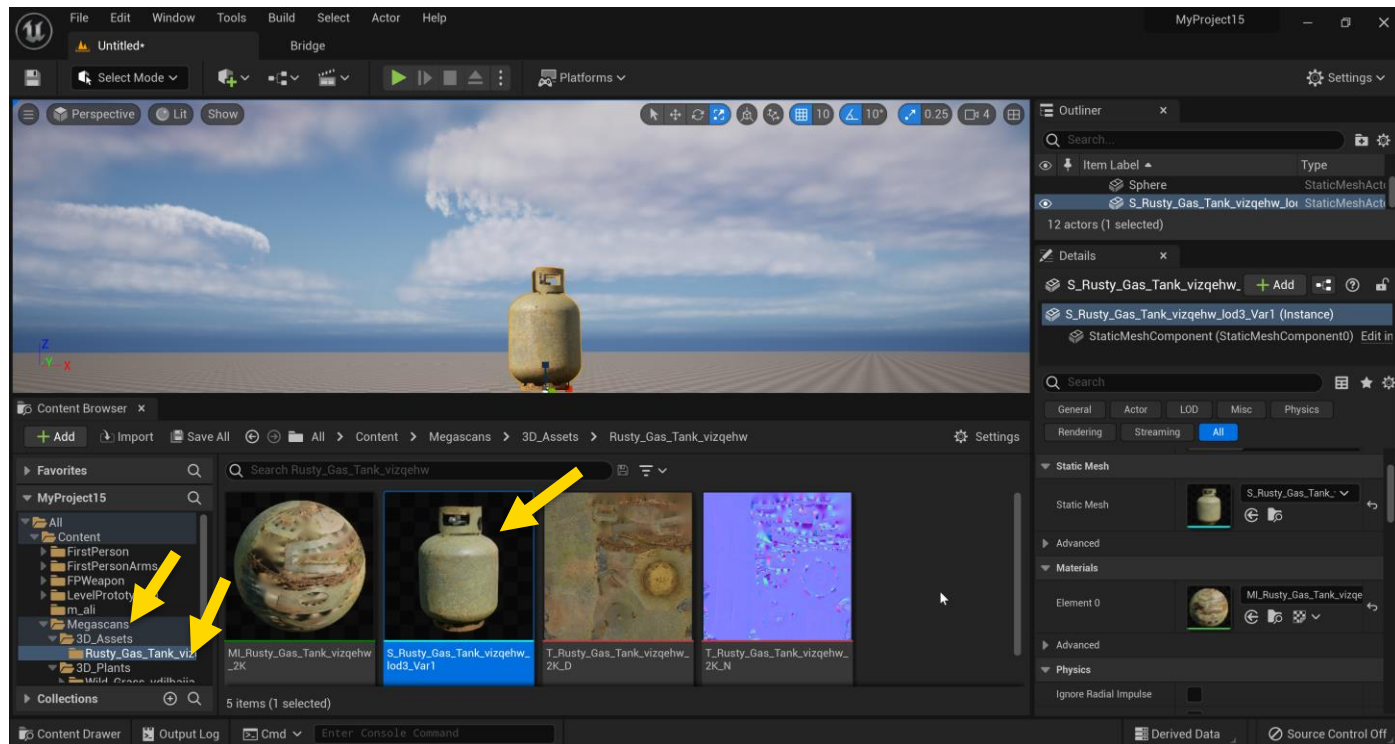
- Click the **Export** button to add the object to the content browser.



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

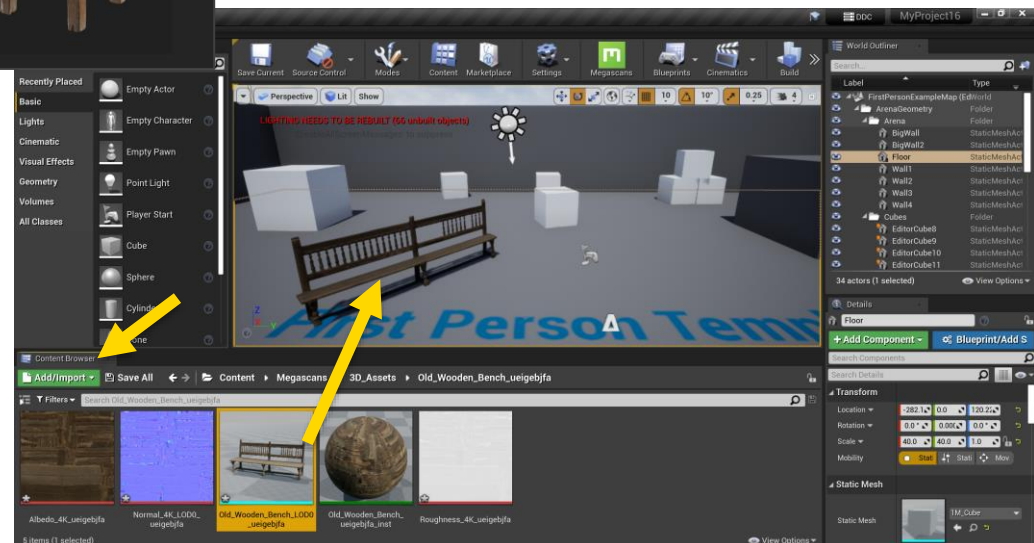
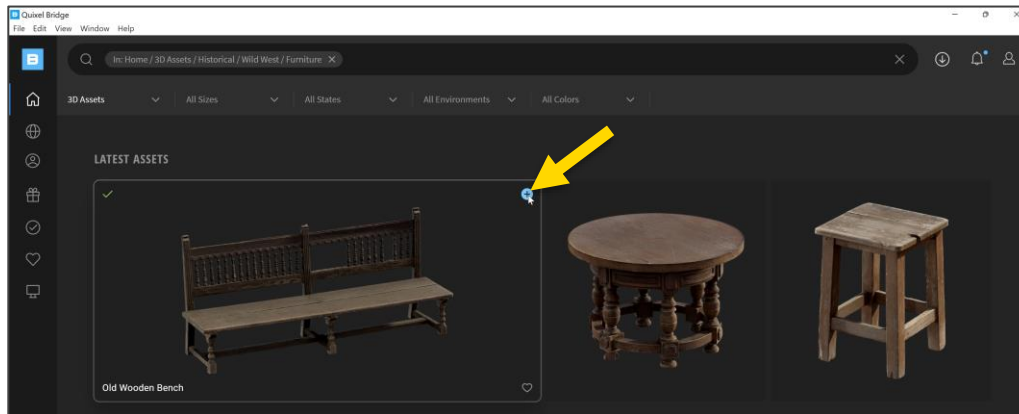
- Megascans > 3D_Assets > Object name



<https://quixel.com/bridge>

Unreal Engine asset store (Quixel Bridge)

- Click the **Export** button to add the object to the content browser.
- Megascans > 3D_Assets > Object name



Lecture activity 3

- Go to Quixel Bridge asset store and find a desired asset and add it into your level.



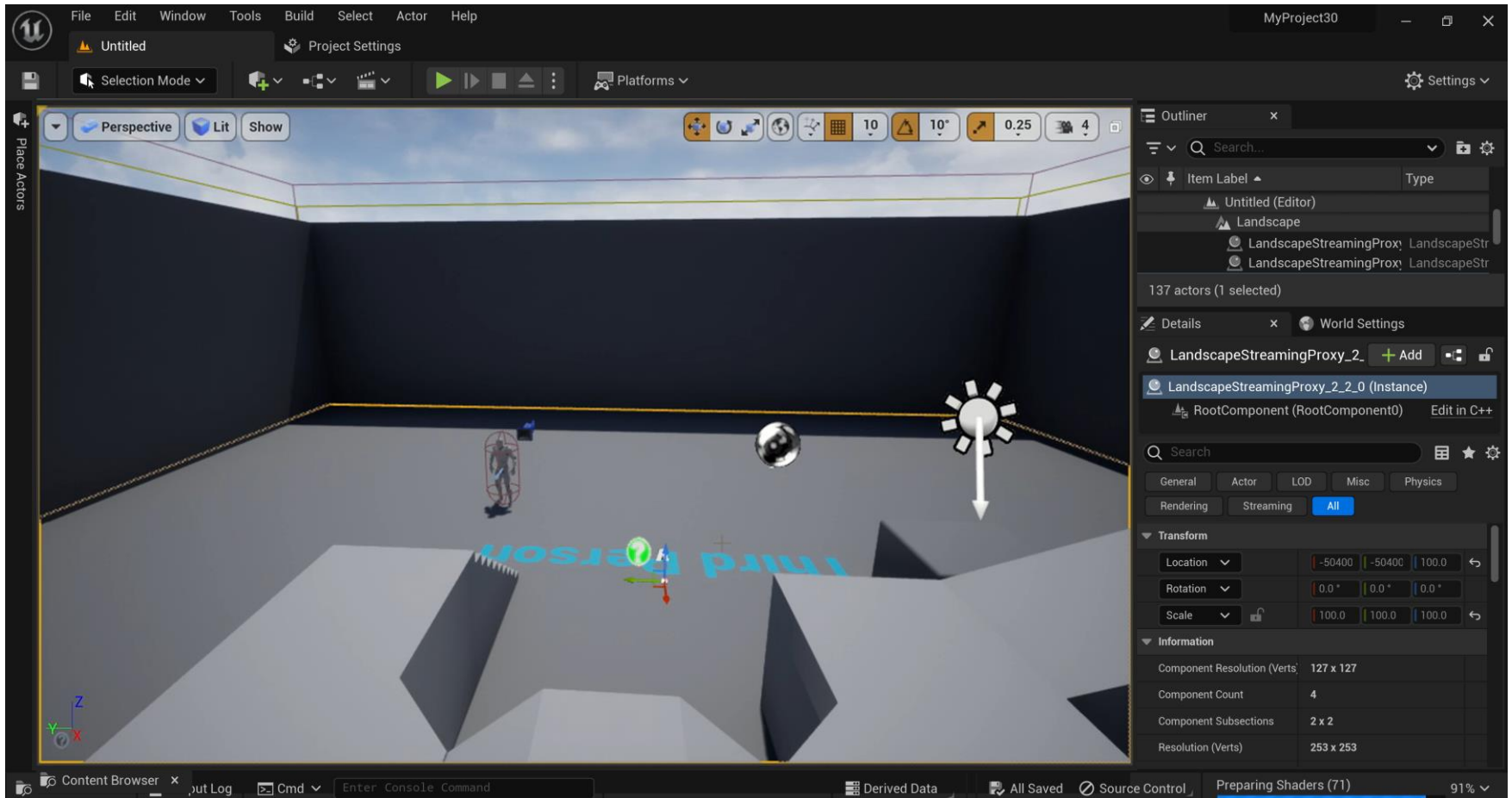
Navigation Controls

Control	Action
Left click + Drag	Moves the camera forward and backward and rotates left and right.
Right click + Drag	Rotates the viewport camera.
Left click + Right click + Drag	Moves up, down, left and right.
Alt + Left click	Orbit or Rotate around the objects.

Navigation Controls

Control	Action
Left click + W / Numpad8 / Up	Moves the camera forward.
Left click + S / Numpad2 / Down	Moves the camera backward.
Left click + A / Numpad4 / Left	Moves the camera left.
Left click + D / Numpad6 / Right	Moves the camera right.
Left click + E / Numpad9 / Page Up	Moves the camera up.
Left click + Q / Numpad7 / Page Down	Moves the camera down.
Left click + Z / Numpad1	Zooms the camera out (raises FOV).
Left click + C / Numpad3	Zooms the camera in (lowers FOV).

Navigation Controls



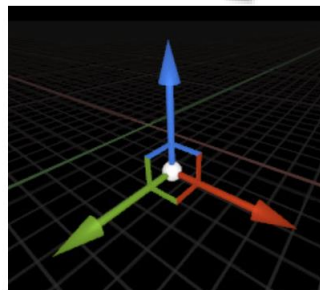
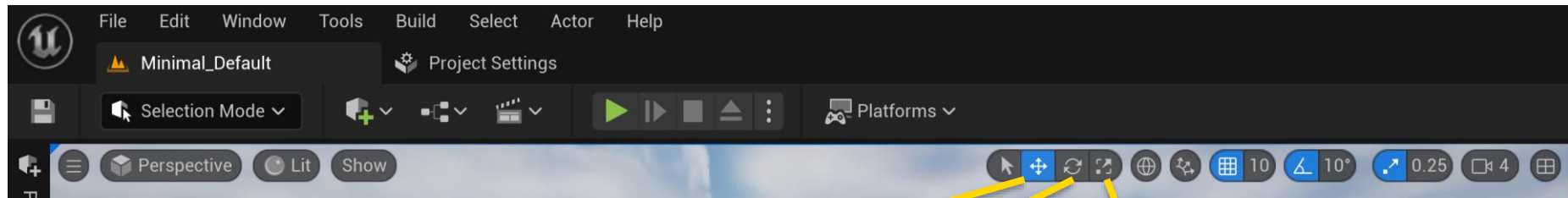
Lecture activity 4

- Navigate your scene using mouse and keyboard keys

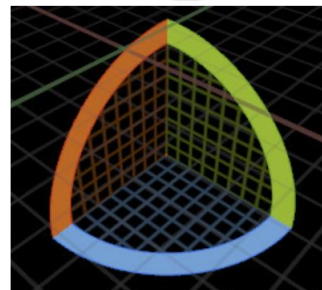


Transforming the objects in the viewport

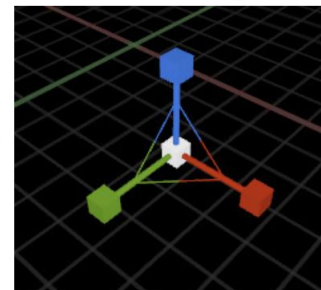
These controls relate to moving, rotating, and scaling actors in the viewports.



Move Tool (W)



Rotate Tool (E)



Scale Tool (R)

Move the objects



Rotate objects

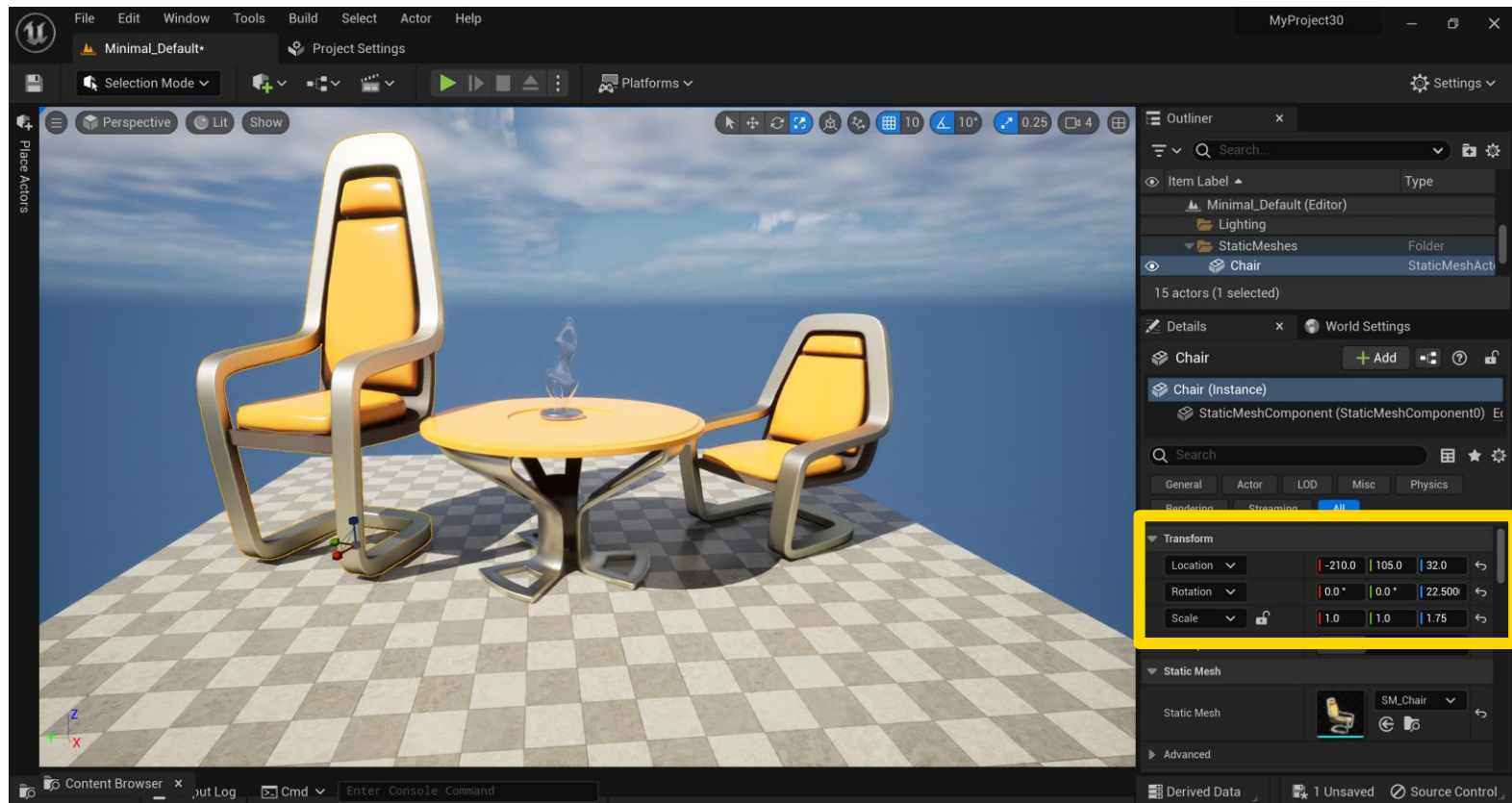


Resize objects



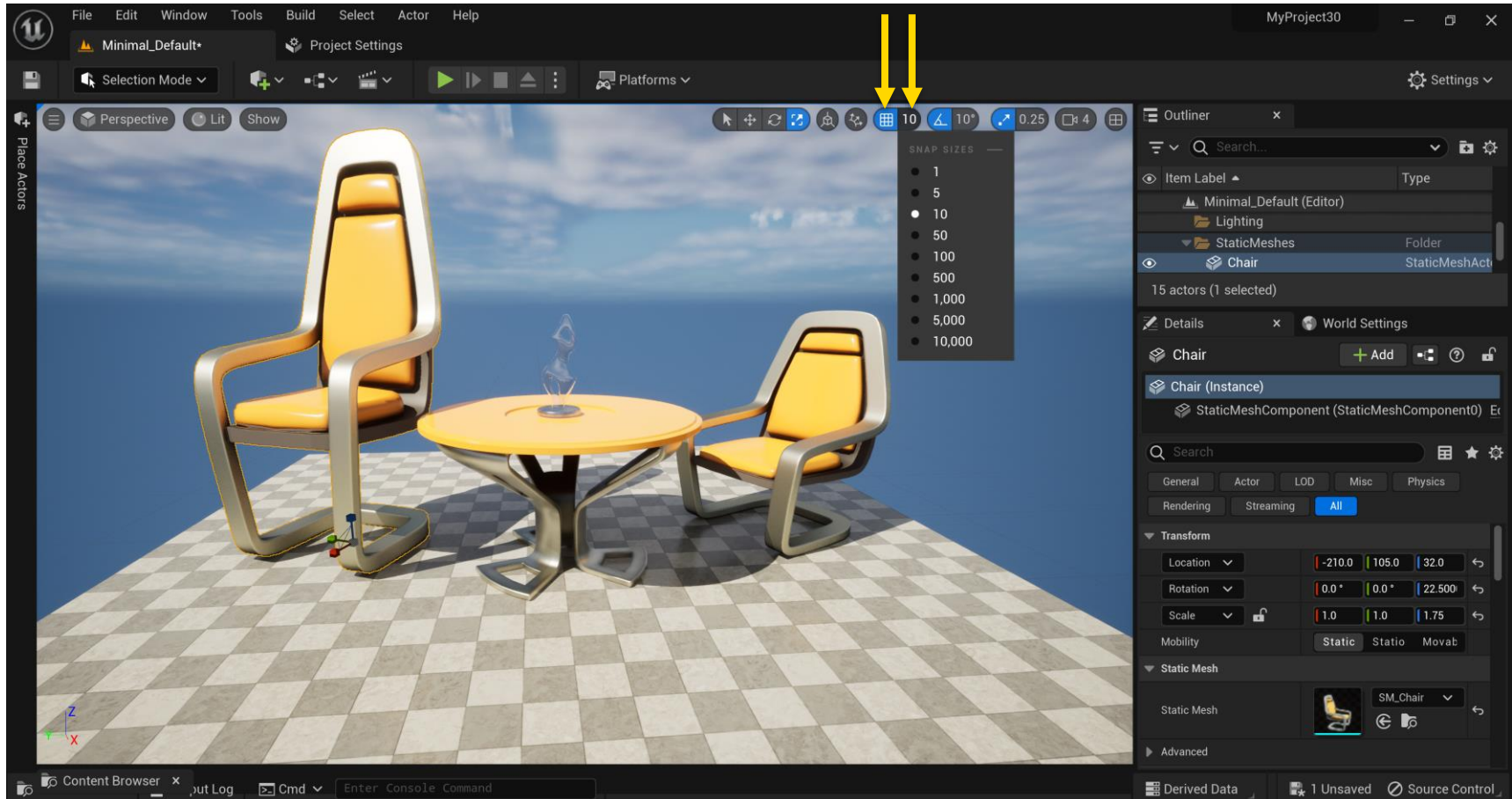
Transforming using the details panel

In the transform section, one unreal unit is equal to one centimeter.



Changing the grid size

You can change the grid size of the Unreal Engine.

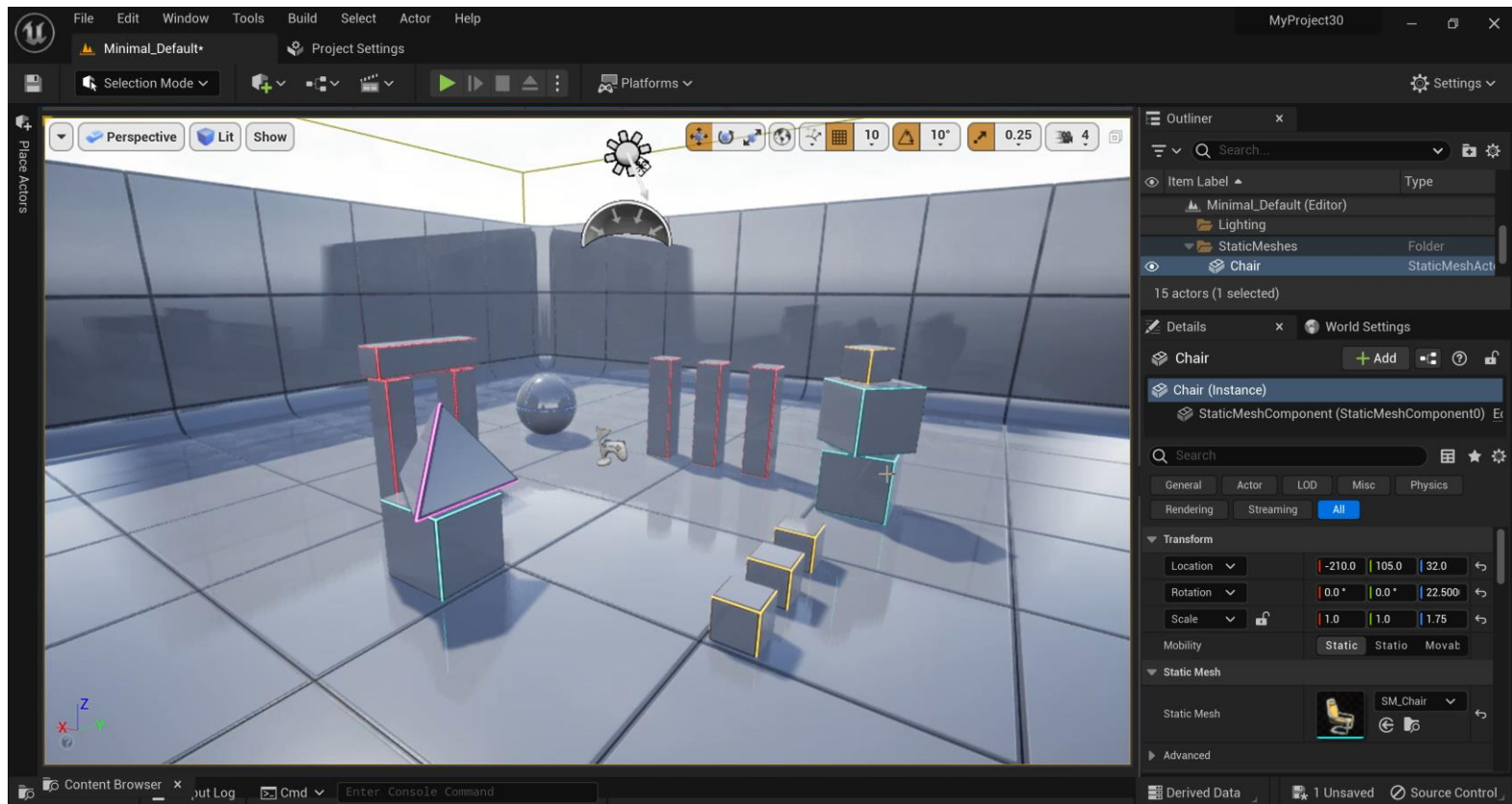


Duplicate an actor (object)

- An Actor is any object that can be placed into a level, such as a Camera, static mesh, or player start location.
- Actors support 3D transformations such as translation, rotation, and scaling.
- Actors can be created (spawned) and destroyed through gameplay code (C++ or Blueprints).

Duplicate an actor (object)

Hold down left **Alt** key on the keyboard and drag the gizmo.



Lecture activity 5

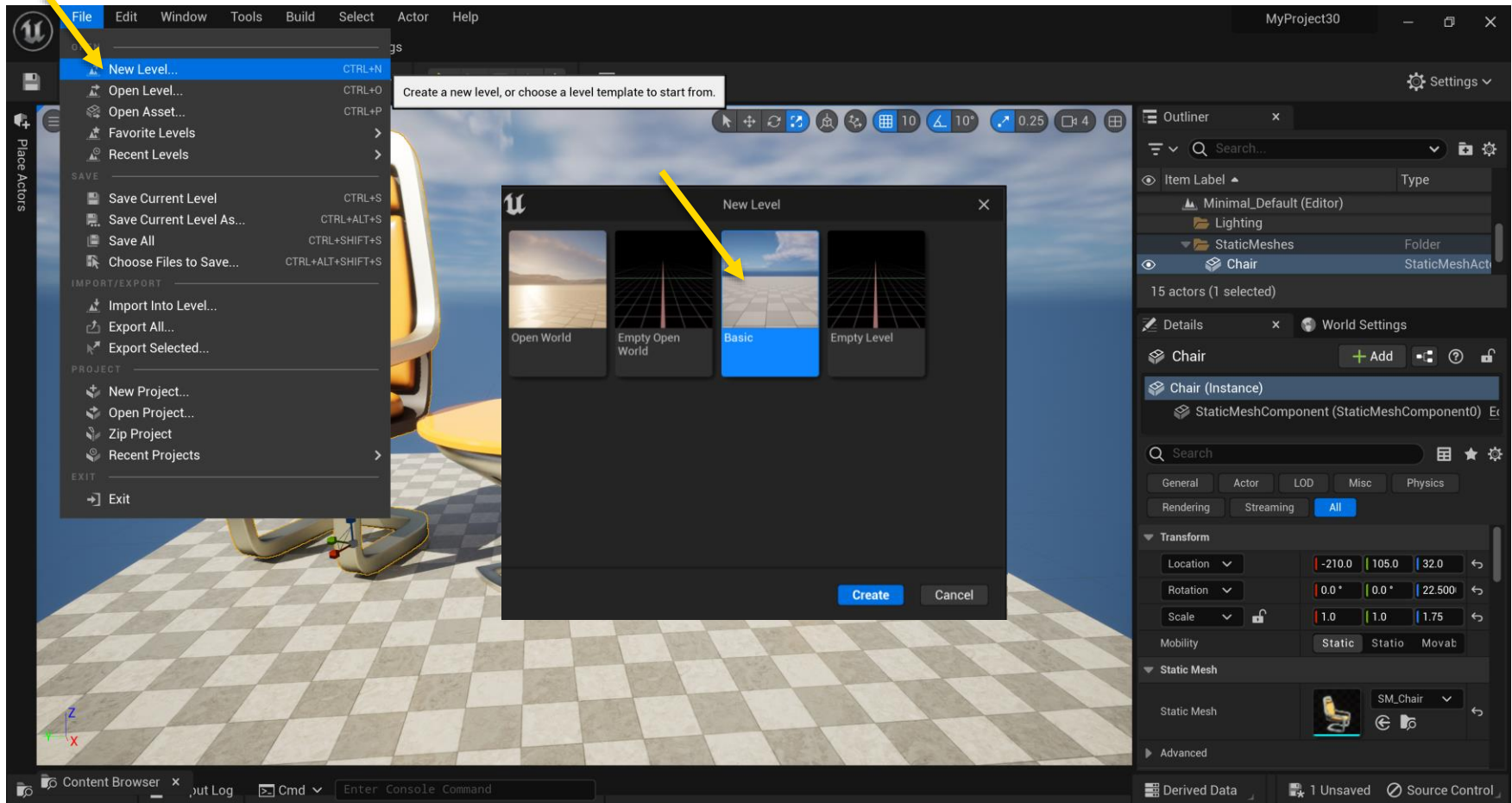
Create a scene with the characteristics below:

- 3 tables
- Each table needs to be on its own floor.
- Each table needs to be decorated with its own statue.
- Each table has unique requests:
 - Table 1, five chairs scale: 0.5
 - Table 2, three chairs scale 0.83
 - Table 3, five chairs scale: 1.6



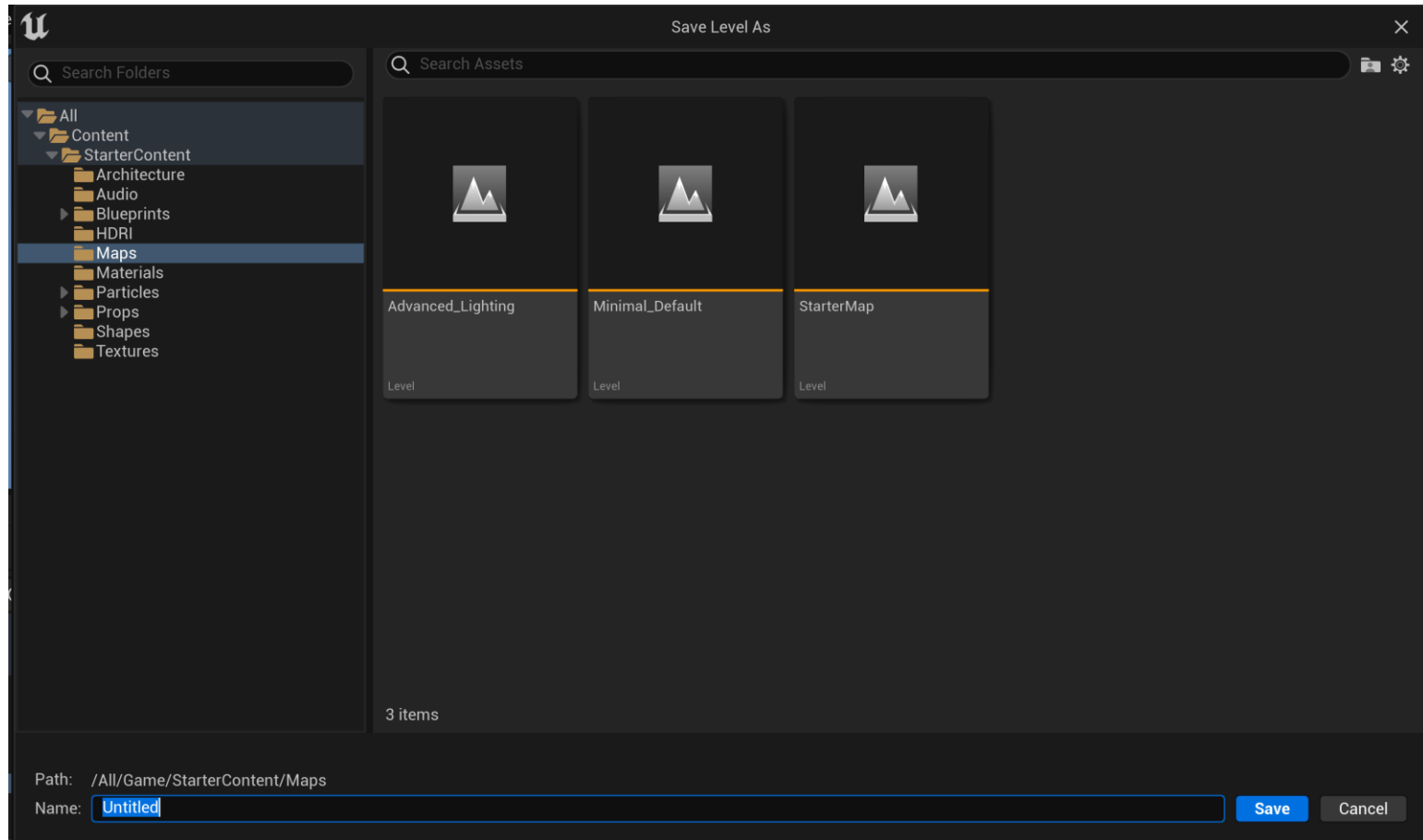
Create a new level

File menu > New Level > Basic



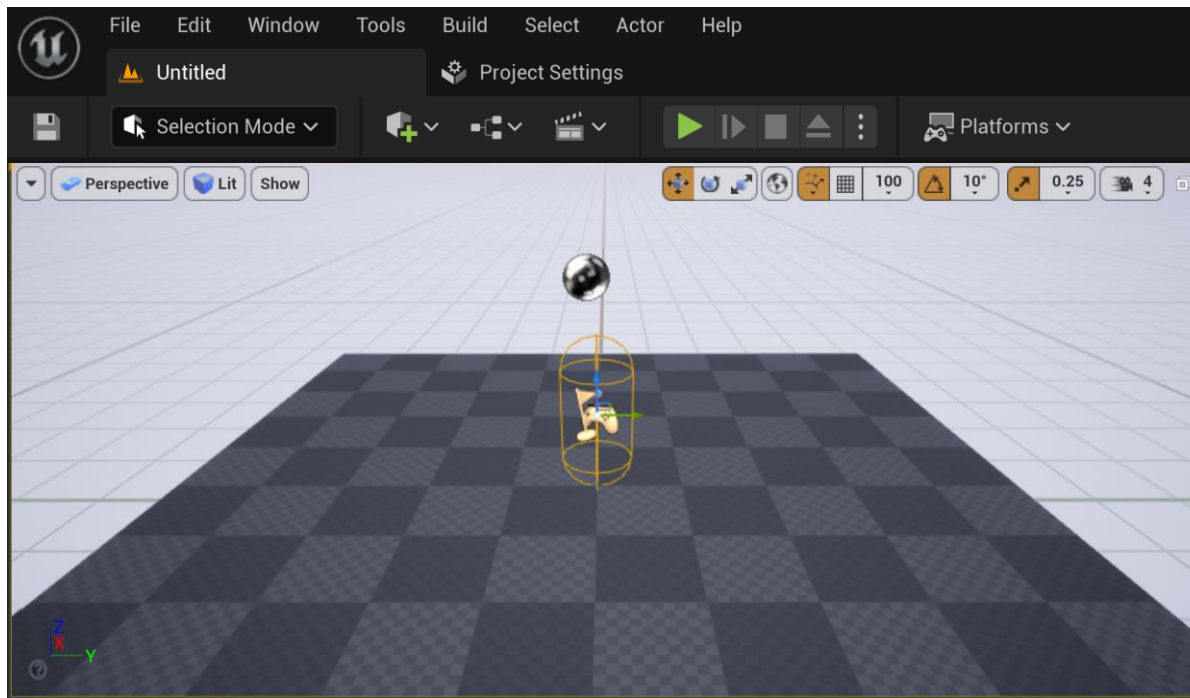
Create a new level

After creating a new level, you should save it into your assets library.



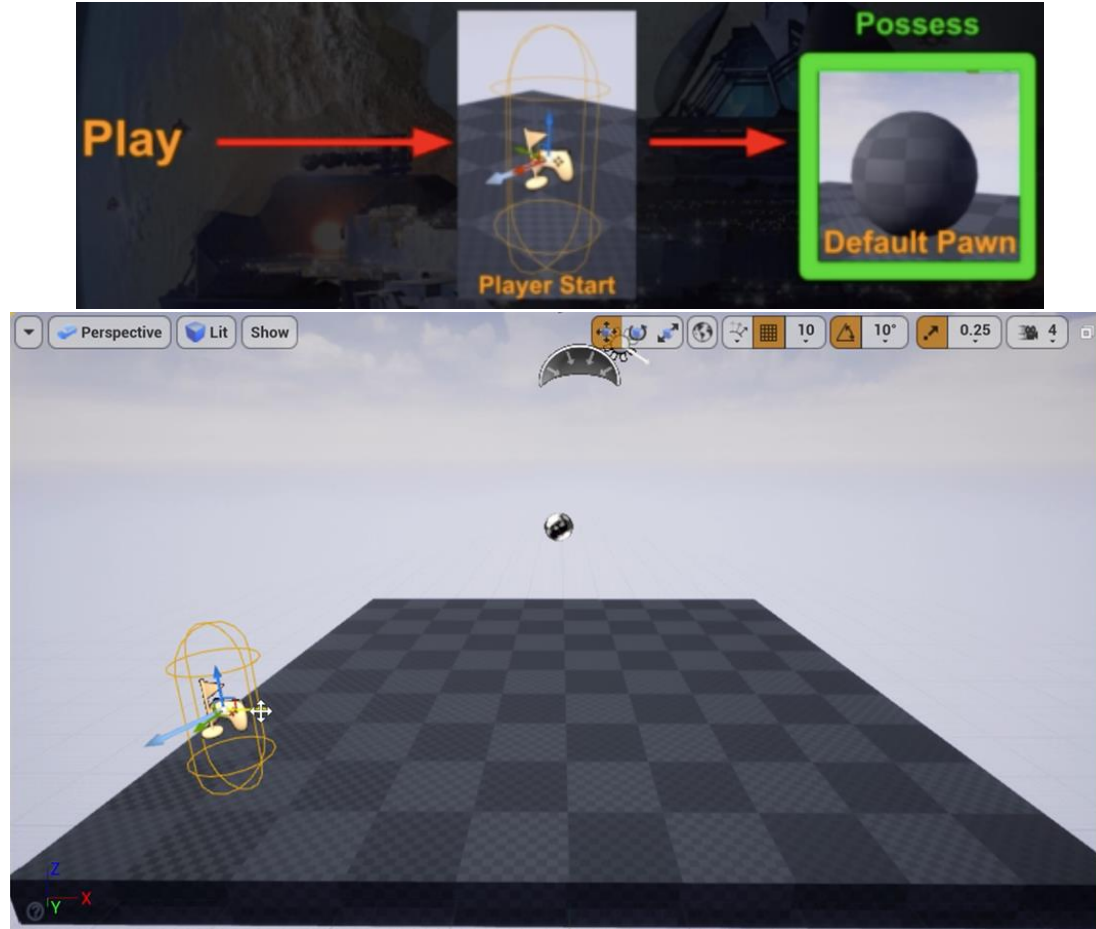
Pawn and player start position

- A pawn is an actor that can be possessed and controlled by a player.
- When you run your game, you see the world game from the lens of the default pawn of the game.



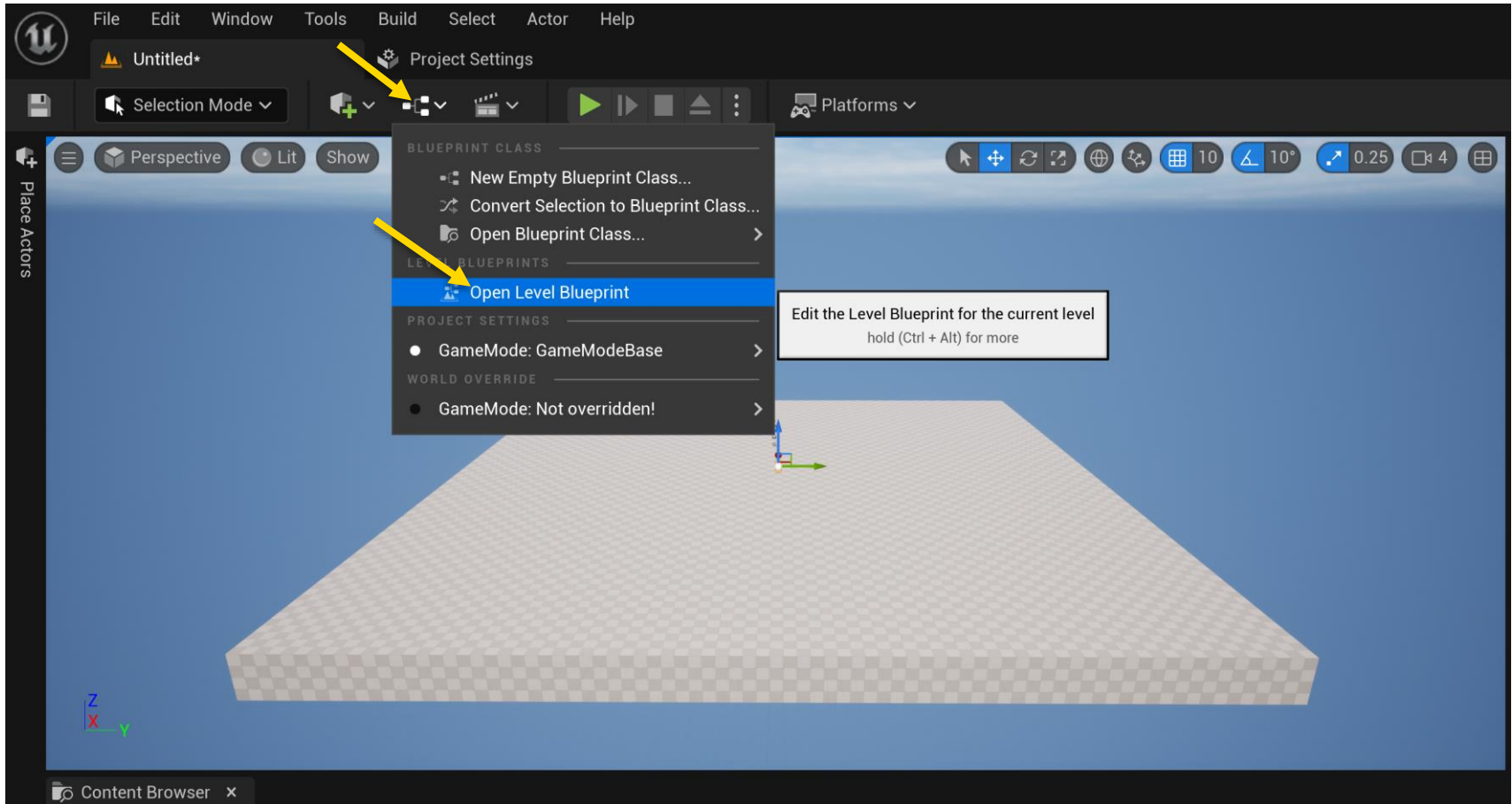
Pawn and player start position

- Use the transform tools to move the starting position and default pawn to a new location.



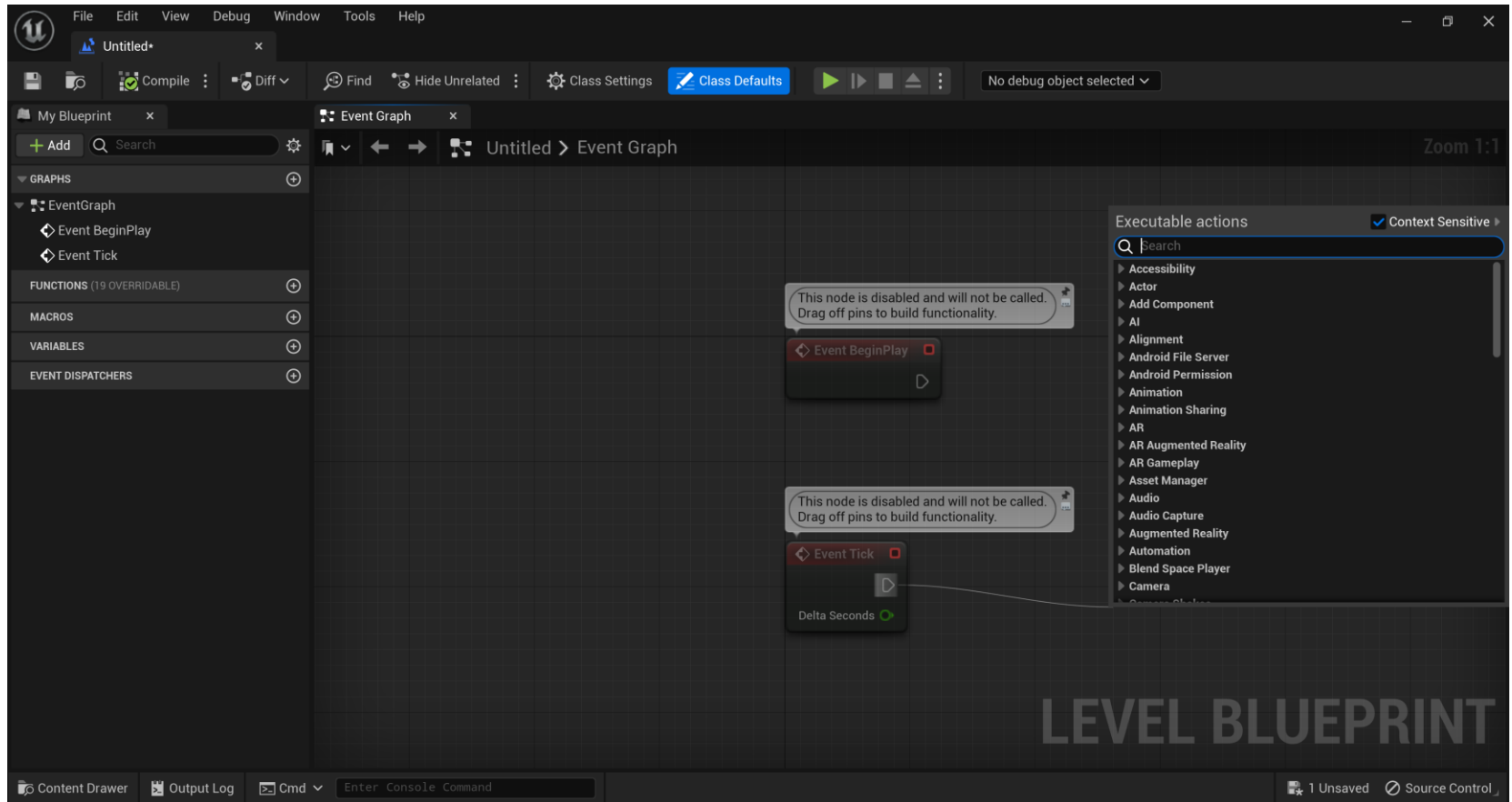
Game scene and blueprint introduction

- In order to add a blueprints script to your level > Click the Blueprints button > Click Open Level Blueprint.



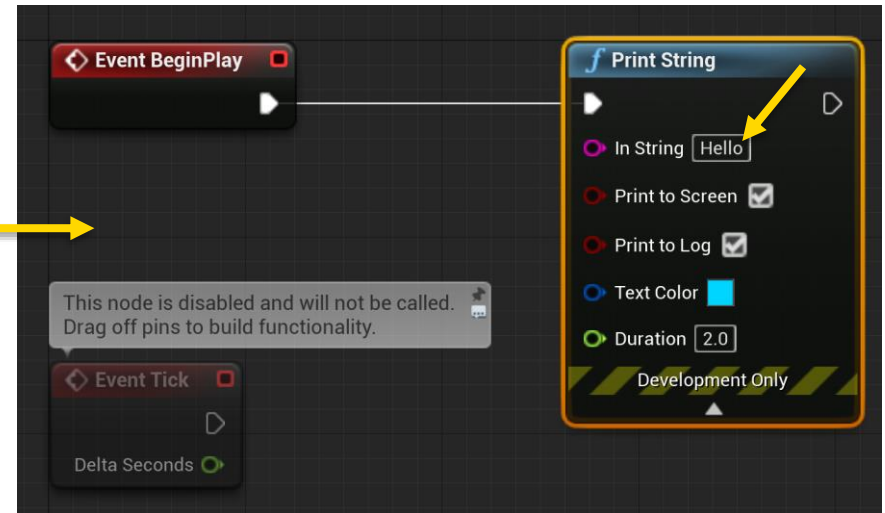
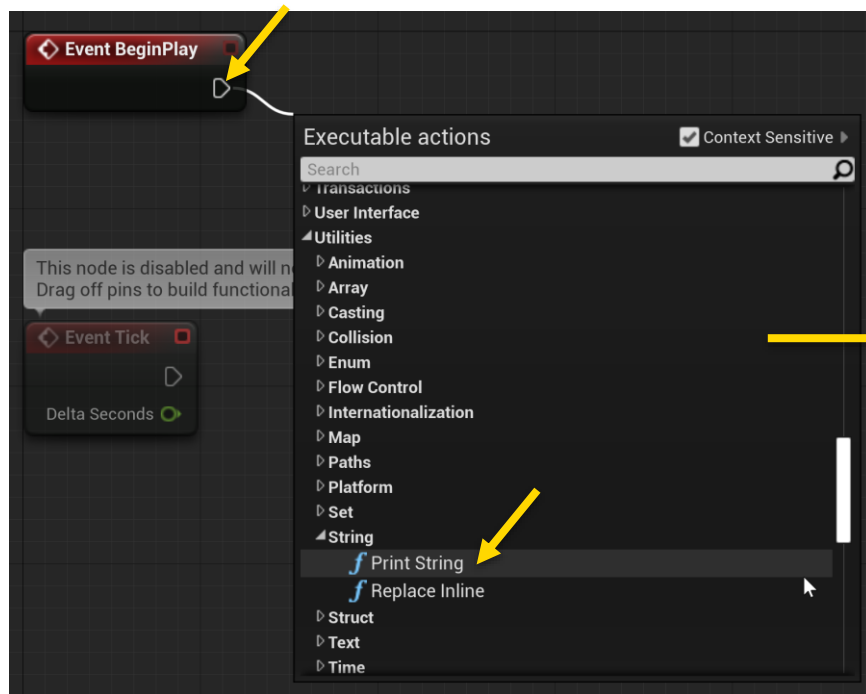
Level blueprints

Drag your mouse from the desired event > Select your desired function or command > Enter values



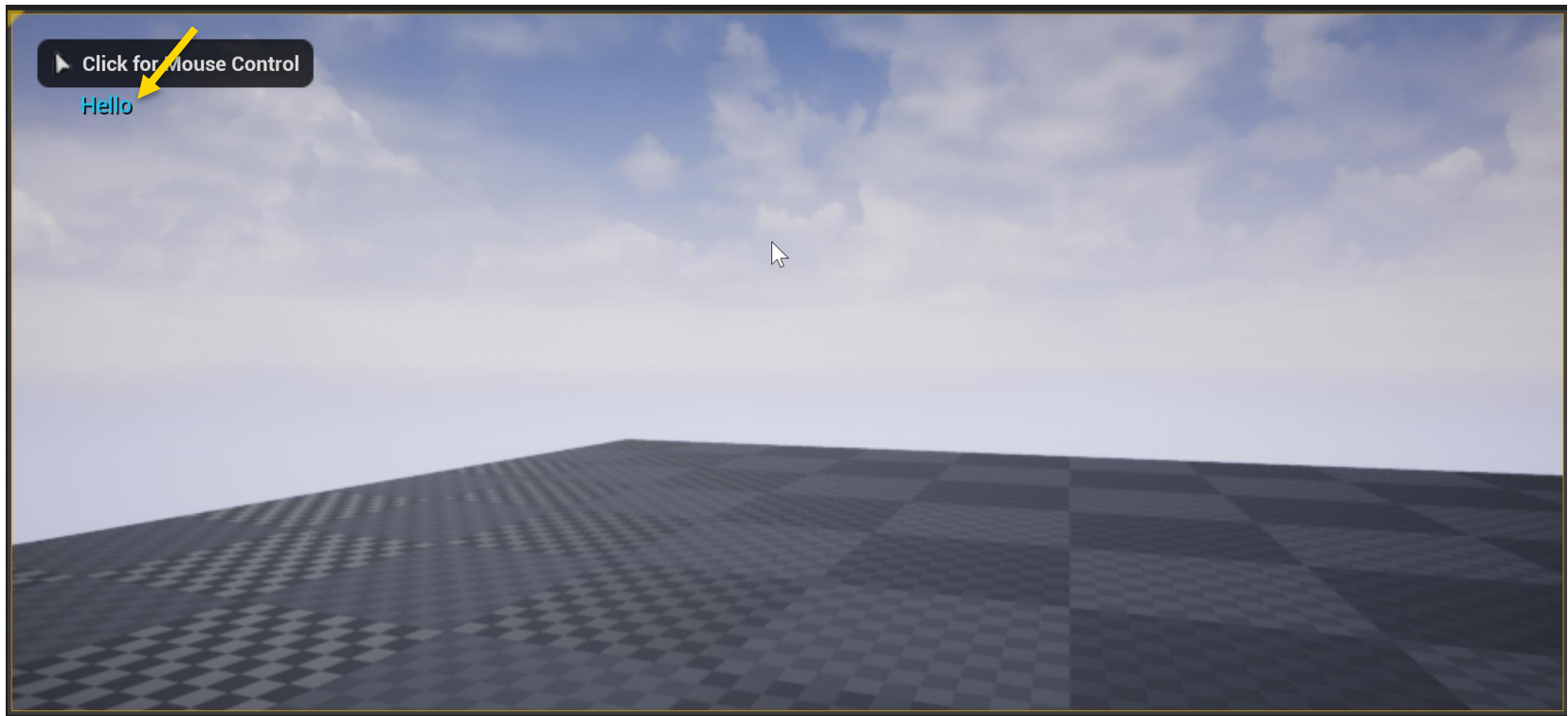
Level blueprints

- When we click and drag from a node, it opens up a window where we can select an executable action.
- To print a message to the screen, use a function called 'print string'.



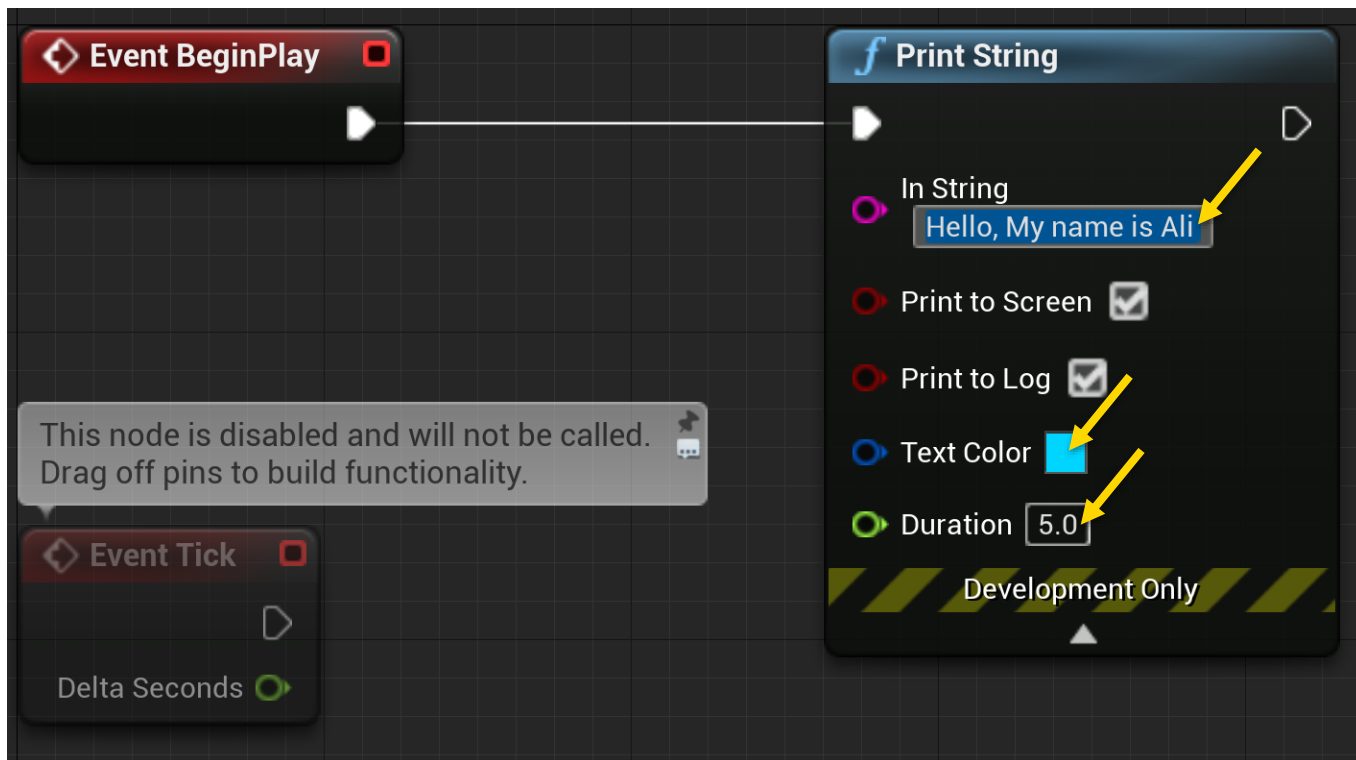
Level blueprints

By running the game, unreal will show a 'Hello' message on the screen.



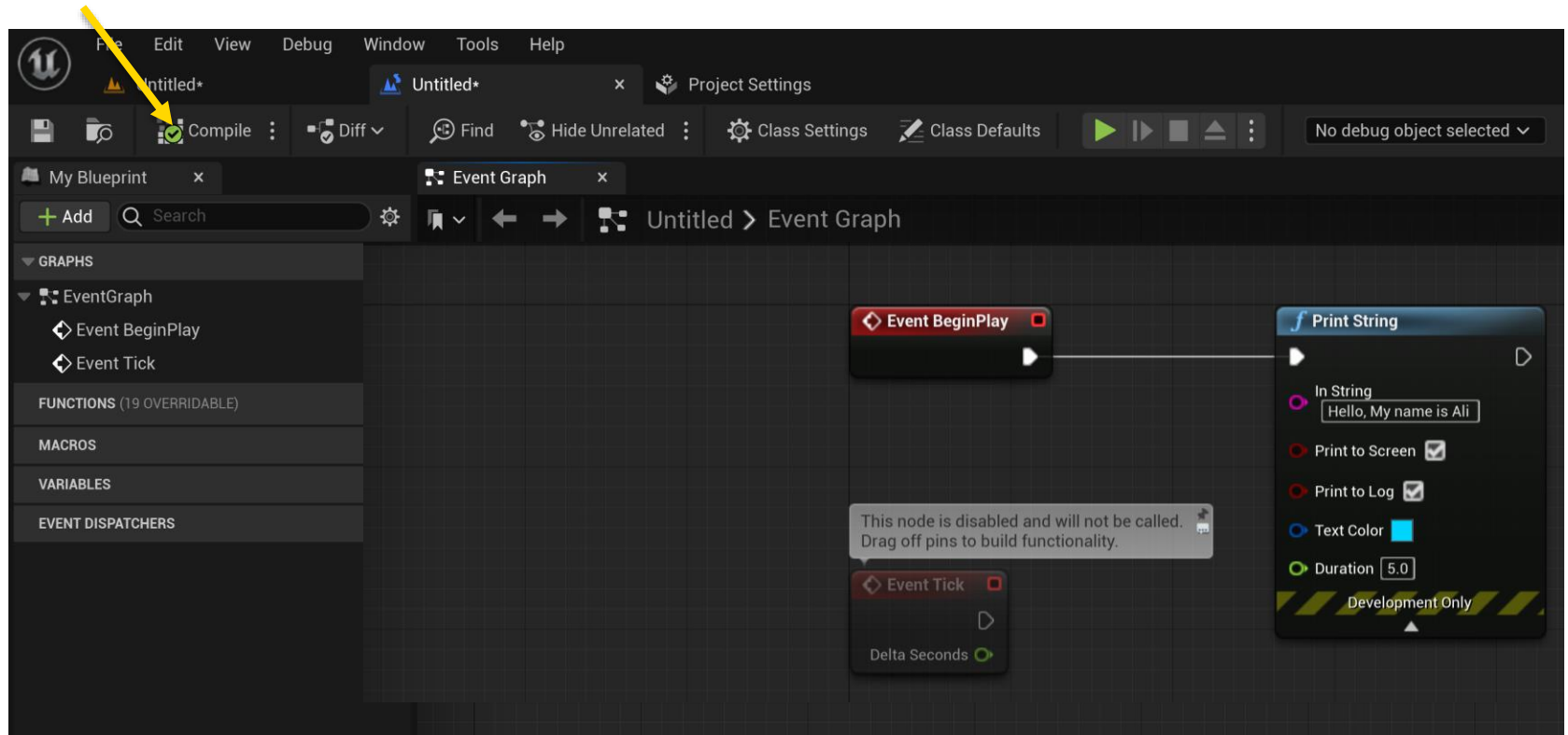
Level blueprints

You can modify the message text and its settings using the Print function attributes.



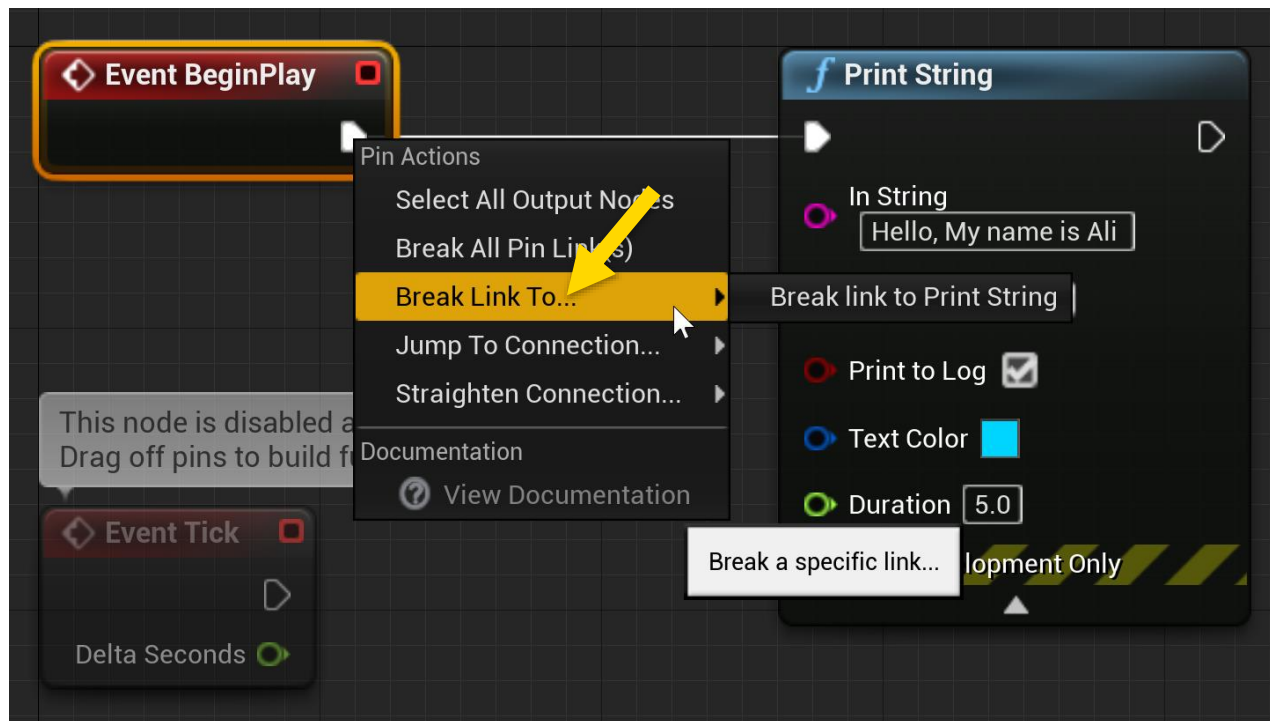
Level blueprints

To examine whether your code is correct, you can press the compile button.



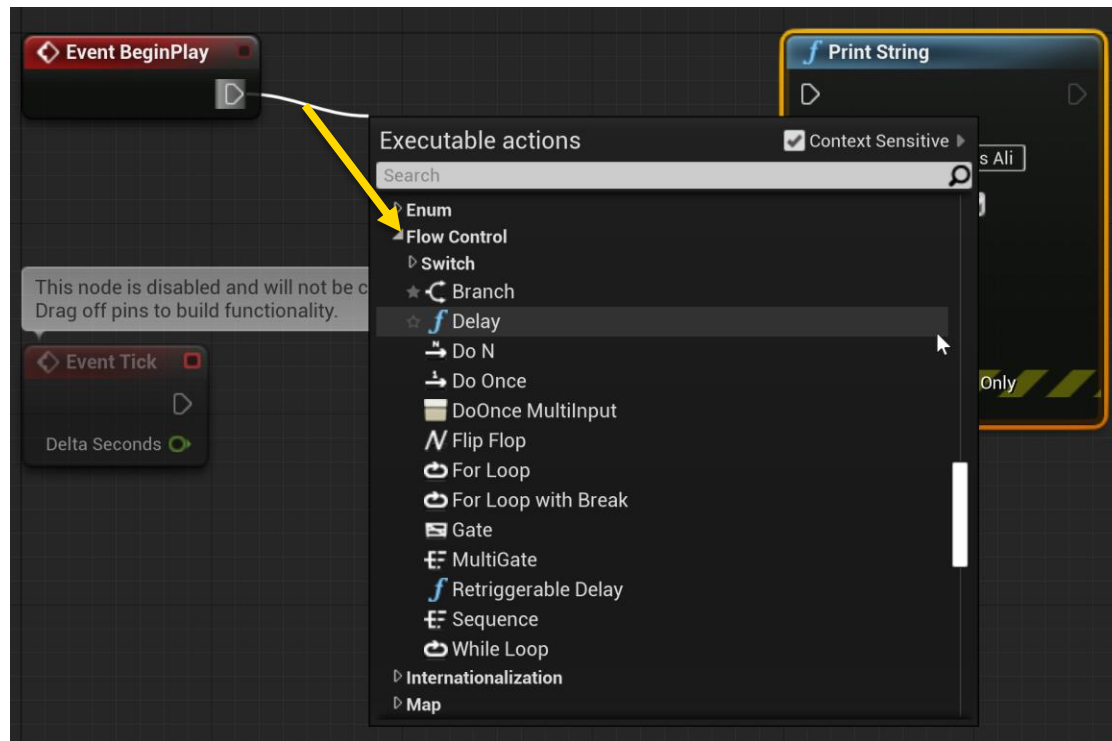
Level blueprints

In order to stop executing a command, right click on any of the execution pins and hit the **Break link to...**



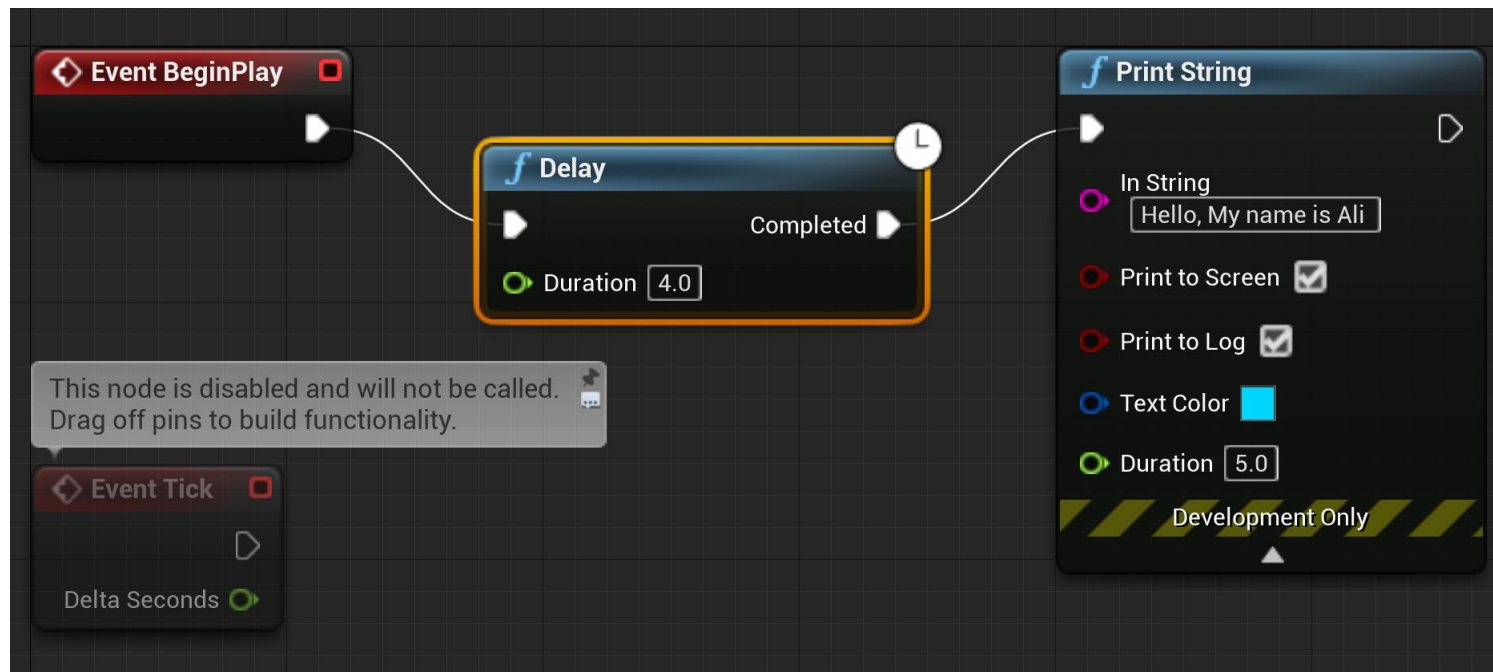
Level blueprints (Changing the code flow)

To change your code flow, use **Utilities** category > Add a **Flow control** node



Level blueprints

- Delay function can be used to run a code with a pause.
- The hello message will be shown after 4 seconds.



Lecture activity 6

- Show different messages on the screen
- Use the delay function to make an interval between showing the messages

